Paladin™ Scan
Modelocked Laser System

Operator's Manual

COHERENT®
Superior Reliability & Performance

Operator's Manual
Paladin™ Scan
Modelocked Laser System

COHERENT.
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Patents referenced in this manual are active when the manual is printed (see last page for date). You are advised to check to see if the patents are still active **http://portal.uspto.gov/external/portal/pair**.

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In the U.S.:

Should you experience any difficulties with your laser or need any technical information, please visit our Web site **www.Coherent.com**. Should you need further assistance, please contact Coherent Technical Support via e-mail **Product.Support@Coherent.com** or telephone, 1-800-367-7890 (1-408-764-4557 outside the U.S.). Please be ready to provide model and laser head serial number of your laser system as well as the description of the problem and any corrective steps attempted to the support engineer responding to your request.

Telephone coverage is available Monday through Friday (except U.S. holidays and company shutdowns). Inquiries received outside normal office hours will be documented by our automatic answering system and will be promptly returned the next business day.

Outside the U.S.:

If you are located outside the U.S., please visit **www.Coherent.com** for technical assistance, or phone your local Service Representative. Service Representative phone numbers and addresses can be found on the Coherent web site.

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Signal Words and Symbols in this Manual

This documentation may contain sections in which particular hazards are defined or special attention is drawn to particular conditions. These sections are indicated with signal words in accordance with ANSI Z-535.6 and safety symbols (pictorial hazard alerts) in accordance with ANSI Z-535.3 and ISO 7010.

Signal Words

Four signal words are used in this documentation: **DANGER**, **WARNING**, **CAUTION** and *NOTICE*.

The signal words **DANGER**, **WARNING** and **CAUTION** designate the degree or level of hazard when there is the risk of injury:

DANGER!
Indicates a hazardous situation that, if not avoided, <u>will</u> result in <u>death or serious injury</u>. This signal word is to be limited to the most extreme situations.

WARNING!
Indicates a hazardous situation that, if not avoided, <u>could</u> result in <u>death or serious injury</u>.

CAUTION!
Indicates a hazardous situation that, if not avoided, could result in <u>minor or moderate injury</u>.

The signal word "***NOTICE***" is used when there is the risk of property damage:

NOTICE!
Indicates information considered important, but not hazard-related.

Messages relating to hazards that could result in both personal injury and property damage are considered safety messages and not property damage messages.

Symbols

The signal words **DANGER**, **WARNING**, and **CAUTION** are always emphasized with a safety symbol that indicates a special hazard, regardless of the hazard level:

This symbol is intended to alert the operator to the presence of important operating and maintenance instructions.

This symbol is intended to alert the operator to the danger of exposure to hazardous visible and invisible laser radiation.

This symbol is intended to alert the operator to the presence of dangerous voltages within the product enclosure that may be of sufficient magnitude to constitute a risk of electric shock.

This symbol is intended to alert the operator to the danger of Electro-Static Discharge (ESD) susceptibility.

This symbol is intended to alert the operator to the danger of crushing injury.

This symbol is intended to alert the operator to the danger of a lifting hazard.

Preface

This manual contains user information for the Paladin Scan Modelocked laser system.

NOTICE
Read this manual carefully before operating the laser for the first time. Special attention must be given to the material in "Section One: Laser Safety" on page 1-1, that describes the safety features built into the laser.

DANGER!
Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

Export Control Laws Compliance

It is the policy of Coherent to comply strictly with U.S. export control laws.

Export and re-export of lasers manufactured by Coherent are subject to U.S. Export Administration Regulations, which are administered by the Commerce Department. In addition, shipments of certain components are regulated by the State Department under the International Traffic in Arms Regulations.

The applicable restrictions vary depending on the specific product involved and its destination. In some cases, U.S. law requires that U.S. Government approval be obtained prior to resale, export or re-export of certain articles. When there is uncertainty about the obligations imposed by U.S. law, clarification must be obtained from Coherent or an appropriate U.S. Government agency.

These commodities, technology, or software are subject to European Union export regulations and local laws. Diversion contrary to European Union law prohibited. The use, sale, re-export, or re-transfer directly or indirectly in any prohibited activities are strictly prohibited.

French Translation/ Traduction Française

Avertissements et symboles de sécurité utilisés dans ce manuel

Cette documentation peut contenir des sections dans lesquelles des risques particuliers sont identifiés ou lorsqu'une attention particulière est requise dans des conditions spécifiques. Ces sections sont indiquées par des avertissements de sécurité selon la norme ANSI Z-535.6 et des symboles de sécurité (pictogrammes d'alertes de risque) selon les normes ANSI Z-535.3 et ISO 7010.

Avertissements de sécurité

Quatre avertissements de sécurité sont utilisés dans cette documentation : DANGER, AVERTISSEMENT, ATTENTION et AVIS.

Les avertissements de sécurité DANGER, AVERTISSEMENT et ATTENTION désignent le degré ou le niveau de risque lorsqu'il existe un risque de blessure :

DANGER !
Désigne une situation de risque qui, si elle n'est pas évitée, aura pour conséquence la mort ou des blessures graves. L'avertissement de sécurité doit être limité aux situations les plus extrêmes.

AVERTISSEMENT !
Désigne une situation de risque qui, si elle n'est pas évitée, pourrait avoir pour conséquence la mort ou des blessures graves.

ATTENTION!
Désigne une situation de risque qui, si elle n'est pas évitée, pourrait avoir pour conséquence une blessure mineure ou modérée.

L'avertissement "***AVIS!***" est utilisé lorsqu'il existe un risque d'endommager le bien:

AVIS!
Indique une information considérée comme importante, mais sans lien avec un risque potentiel.

Les messages relatifs à des risques pouvant avoir pour conséquence à la fois des blessures aux personnes ou des dommages matériels sont considérés comme des messages de sécurité et non comme des messages de dommages matériels.

Symboles

Les avertissements de sécurité DANGER, AVERTISSEMENT et ATTENTION sont systématiquement accentués par un symbole de sécurité qui indique un risque particulier, indépendamment du niveau de risque:

Ce symbole est prévu pour alerter l'opérateur de la présence d'instructions importantes d'utilisation et d'entretien.

Ce symbole est prévu pour alerter l'opérateur du danger de l'exposition au rayonnement laser visible et invisible dangereux.

Ce symbole est prévu pour alerter l'opérateur de la présence de tensions dangereuses dans le carter du produit qui peuvent être suffisamment importantes pour constituer un risque de décharge électrique.

Ce symbole est prévu pour alerter l'opérateur du danger de susceptibilité aux décharges électrostatiques (ESD).

Ce symbole est prévu pour alerter l'opérateur du danger de blessure par écrasement.

Ce symbole est prévu pour alerter l'opérateur du danger lié à un risque de levage.

Préface

Ce manuel contient l'information utilisateur pour le système laser Paladin Scan Modelocked.

Veuillez lire attentivement ce manuel avant d'actionner le laser pour la première fois. Une attention particulière doit être portée au matériel dans la " Section 1 : Sécurité du laser " à la page 1, qui décrit les dispositifs de sécurité intégrés au laser.

L'utilisation de commandes ou de réglages ou l'exécution de procédures autres que ceux spécifiés dans la présente peut avoir comme conséquence l'exposition à un rayonnement dangereux.

Conformité avec les lois de contrôle des exportations

Coherent a pour politique de se conformer strictement aux lois de contrôle des exportations des États-Unis.

L'exportation et la réexportation des lasers construits par Coherent sont sujettes aux règlements d'administration des exportations des États-Unis, gérés par le département américain du commerce. En outre, les expéditions de certains composants sont réglementées par le département d'État en vertu de la réglementation visant le trafic international d'armes.

Les restrictions applicables varient selon le produit spécifique impliqué et sa destination. Dans certains cas, la loi des États-Unis exige que l'accord du gouvernement des États-Unis soit obtenu avant la revente, l'exportation ou la réexportation de certains articles. Quand il y a incertitude sur les obligations imposées par la loi des États-Unis, une clarification doit être obtenue auprés de Coherent ou d'un organisme gouvernemental compétent des États-Unis.

Ces produits, technologies et logiciels sont soumis aux règles d'exportation de la Communauté Europeénne et aux lois locales. Des versions contraires aux lois Européennes sont interdites.

L'utilisation, la vente, la ré-exportation ou un transfert directement ou indirectement pour toutes activités interdites sont strictement interdites.

SECTION ONE: LASER SAFETY

Introduction

The Paladin Scan Laser system has been designed to provide protection to the operator in the event of any single component failure, provided that the system is installed and operated properly as described in the Operator's manual. All laser users must read, understand, and follow the safety warnings and operating instructions contained within the operator's manual.

Because the manufacturer is unable to guarantee the safety of laser users in the event of two independent component failures, this equipment must not be operated if there is evidence of any personnel hazard, component failure, improper installation, or significant damage. The failure of a single component is not hazardous, but may allow for no protection against, or warning of, a hazardous condition if another failure occurs. Routinely inspect the laser system for evidence of potential safety hazards or component failures.

If it is suspected that the laser system is missing safety related parts, has been damaged, or may otherwise be unsafe, turn the laser off, disconnect the input power immediately, and contact your local Coherent service representative. Do not operate the laser until all potential safety hazards have been eliminated. Do not remove any housings or protective covers unless directed to do so by a certified Coherent service representative.

This user information is in compliance with the following standards for Light-Emitting Products IEC 60825-1 / EN 60825-1 *"Safety of laser products - Part 1: Equipment classification and requirements"* 21 CFR Title 21 Chapter 1, Subchapter J, Part 1040 *"Performance standards for light-emitting products"*.

WARNING!
LASER RADIATION - AVOID EYE OR SKIN EXPOSURE TO DIRECT OR SCATTERED RADIATION CLASS 4 LASER PRODUCT!

WARNING!
Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

This laser safety section must be reviewed thoroughly prior to operating the Paladin system. Safety instructions presented throughout this manual must be followed carefully.

Hazards

Hazards associated with lasers generally fall into the following categories:

- Biological hazards from exposure to laser radiation that may damage the eyes or skin
- Electrical hazards generated in the laser power supply or associated circuits
- Chemical hazards resulting from contact of the laser beam with volatile or flammable substances, or released as a result of laser material processing

The above list is not intended to be exhaustive. Anyone operating the laser must consider the interaction of the laser system with its specific working environment to identify potential hazards.

Optical Safety

Laser light, because of its special properties, poses safety hazards not associated with light from conventional sources. The safe use of lasers requires that all laser users, and everyone near the laser system, are aware of the dangers involved. The safe use of the laser depends upon the user being familiar with the instrument and the properties of coherent, intense beams of light.

The safety precautions listed below are to be read and observed by anyone working with or near the laser. At all times, ensure that all personnel who operate, maintain or service the laser are protected from accidental or unnecessary exposure to laser radiation exceeding the accessible emission limits defined in the laser safety standards.

WARNING!
Direct eye contact with the output beam from the laser may cause serious eye injury and possible blindness.

The greatest concern when using a laser is eye safety. In addition to the main beam, there are often many smaller beams present at various angles near the laser system. These beams are formed by specular reflections of the main beam at polished surfaces such as lenses or beamsplitters. While weaker than the main beam, such beams may still be sufficiently intense to cause eye damage.

Laser beams are powerful enough to burn skin, clothing, or combustible materials, even at some distance. They can ignite volatile substances such as alcohol, gasoline, ether, and other solvents, and can damage light-sensitive elements in video cameras, photomultipliers, and photodiodes. The user is advised to follow the control measures below.

Recommended Precautions and Guidelines

1. Observe all safety precautions in the preinstallation and operator's manuals.
2. Always wear appropriate eyewear for protection against the specific wavelengths and laser energy being generated.
3. Always protect skin against specific wavelengths and laser energy being generated.
4. Avoid wearing watches, jewelry, or other objects that may reflect or scatter the laser beam.
5. Stay aware of the laser beam path, particularly when external optics are used to steer the beam.
6. Provide enclosures for beam paths whenever possible.
7. Use appropriate energy-absorbing targets for beam blocking.
8. Block the beam before applying tools such as Allen wrenches or ball drivers to external optics.
9. Limit access to the laser to trained and qualified users who are familiar with laser safety practices. When not in use, lasers should be shut down completely and made off-limits to unauthorized personnel.
10. Terminate the laser beam with a light-absorbing material. Laser light can remain collimated over long distances and therefore presents a potential hazard if not confined. It is good practice to operate the laser in an enclosed room.

11. Post laser warning signs in the area of the laser beam to alert those present.
12. Exercise extreme caution when using solvents in the area of the laser.
13. Never look directly into the laser light source or at scattered laser light from any reflective surface, even when wearing laser safety eyewear. Never sight down the beam.
14. Set up the laser so that the beam height is either well below or well above eye level.
15. Avoid direct exposure to the laser light. Laser beams can easily cause flesh burns or ignite clothing. Wear suitable clothing for the wavelength being generated to protect skin against laser radiation.
16. Advise all those working with or near the laser of these precautions.

CAUTION!
Laser safety eyewear protects the user from accidental exposure to laser radiation by blocking light at the laser wavelengths. However, laser safety eyewear may also prevent the operator from seeing the beam or the beam spot. Exercise extreme caution even while wearing safety glasses.

Electrical Safety

The Paladin Scan Laser system contains potentially hazardous voltages inside the protective enclosures of the power supply and laser head. Do not open the protective enclosures. Do not operate the laser system without a protective earth (safety) ground connected to the facility power inlet connector.

Before operating the application, an overall risk assessment and an evaluation of the application is necessary.

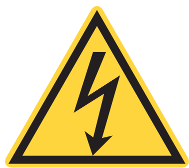
This equipment is for indoor use (air-conditioned clean room only) in non-hazardous locations, operated by qualified personnel skilled in its use.

The main power cord shall comply with the National Standards and/or Electrical Codes of the country in question. Only earthed mains power cord shall be used.

The main power supply connection cord shall be rated for the maximum equipment current and not longer than 3 m.

The disconnecting device shall be located electrically as close as practicable to the supply. Power-consuming components shall not be electrically located between the supply source and the disconnecting device, except that electromagnetic interference suppression circuits are permitted to be located on the supply side of the disconnecting device.

Overcurrent protection devices as required per installation instructions shall be provided at final installation.



DANGER!
Normal operation of the Paladin should not require access to the power supply circuitry. Removing the power supply cover will expose the user to potentially lethal electrical hazards. Contact an authorized service representative before attempting to correct any problem with the power supply.

Recommended Precautions and Guidelines

The following precautions must be observed by everyone when working with potentially hazardous electrical circuitry:

DANGER!
When working with electrical power systems, the rules for electrical safety must be strictly followed. Failure to do so could result in the exposure to lethal levels of electricity.

1. Disconnect main power lines before working on any electrical equipment when it is not necessary for the equipment to be operating.
2. Do not short or ground the power supply output. Protection against possible hazards requires proper connection of the ground terminal on the power cable, and an adequate external ground. Check these connections at the time of installation, and periodically thereafter.
3. Never work on electrical equipment unless there is another person nearby who is familiar with the operation and hazards of the equipment, and who is competent to administer first aid.
4. When possible, keep one hand away from the equipment to reduce the danger of current flowing through the body if a live circuit is touched accidentally.

5. Always use approved, insulated tools.
6. Special measurement techniques are required for this system. A technician who has a complete understanding of the system operation and associated electronics must select ground references.

Safety Features and Compliance to Government Regulations

The following features are incorporated into the Paladin laser system to conform to several government requirements. The applicable United States Government requirements are contained in 21 CFR, subchapter J, part II administered by the Center for Devices and Radiological Health (CDRH). The European Community requirements for product safety are specified in the Low Voltage Directive (LVD) (2014/35/EU). The Low Voltage Directive requires that lasers comply with the standard UL 61010-1:2012 and CAN/CSA-C22.2 No. 61010-1-12 and EN60825-1 “Radiation Safety of Laser Products”. Compliance of this laser with the (LVD) requirements is certified by the CE mark.

Laser Classification

The governmental standards and requirements specify that the laser must be classified according to the output power or energy and the laser wavelength. The Paladin is classified as Class IV based on 21 CFR, subchapter J, part II, section 1040-10 (d). According to the European Community standards, the Paladin laser is classified as Class 4 based on EN 60825-1, clause 9. In this manual, the classification will be referred to as Class 4.

Protective Housing

The laser head is enclosed in a protective housing that prevents human access to radiation in excess of the limits of Class radiation as specified in the 21CFR, Part 1040 Section 1040.10 (f)(1) and EN 60825-1/IEC 60825-1 Clause 6.2 except for the output beam, which is Class 4.

Shutter Interlock

The shutter interlock is located at the rear of the power supply (See “EMBS” in Figure 4-2). The shutter will not open unless the shutter interlock circuit is closed. If the shutter interlock circuit is opened while the laser is operating, it will cause the shutter to close but will not turn the laser off.

For the EMBS, a short between the top two pins allows the shutter to operate normally while an open prevents the shutter from being opened. Impedance of the short must be less than 100 Ohms. Impedance of the open must be greater than 5 K Ohms, galvanic isolation preferred. Current through a short will typically be 20 mA.

Shutter Operation

The EMBS (Electro-Magnetic Beam Shutter) is fail-safe related to EN ISO 13849-1 category 2. It is normally closed and closes in direction of gravity when the head is positioned with its feet toward the ground. The shutter status is monitored continuously via two light barriers, one for the off status and one for on status.

The shutter will close in standard operation within 10 ms. The reaction time of the shutter at an alarm is 25 ms.

NOTICE
The shutter is a required safety feature for laser operation. It is not to be used as a means to control exposure time!

External Interlock

The system will not operate with the interlock open. An EXTERNAL INTERLOCK connector is located at power supply rear panel (See “Interlock” in Figure 4-2). The interlock status is monitored by the CPU of the control board located in the master power supply. If the interlock is opened, a message will be displayed on the power supply front panel.

To properly integrate the laser in the safety control of the application, an external interlock circuit must be connected to the laser system and wired to a door switch or access panel to provide additional operating safety. When a door is opened, the laser will shut down and the shutter will close. The fault must be cleared and the diodes turned on to restart the laser.

To incorporate an external safety interlock circuit into the laser system, turn off the laser and remove the jumper from the interlock connector on the master power supply rear panel. Attach a user furnished external interlock circuit to this connector. Any external interlock circuit must be equivalent to a mechanical closure of the circuit. Under no circumstances should an external voltage or current source be connected to this circuit. External interlock circuitry must be isolated from all other electrical circuit or grounds.

A disruption in the interlock circuit shuts down the laser diodes and eliminates all hazardous light emission. The interlock loop is established as 20 mA current loop, that must be closed to enable laser radiation.

The integrator must ensure the correct wiring of the system. This includes the protection of the wiring at power input, interlock loop and EMBS against a short circuit or cross circuit. The integrator, must avoid any supply to the control inputs (interlock loop, EMBS) of the system.

The "Shutter Interlock" and "External Interlock" inputs of the power supply must be connected to EMERGENCY STOP and/or EMERGENCY OFF circuit of the final application as required at the risk assessment of the final application.

Laser Emission Indicators

The appropriately labeled lights on both the power supply, and the laser head illuminate when the laser on button is set to the LASER ON position before laser emission can occur. White or yellow lights are used and are visible while wearing safety glasses.

Operating Controls

The laser controls are positioned so that the operator is not exposed to laser emission while manipulating the controls.

DANGER!
Use of the system controls in a manner other than that described herein may impair the protection provided by the system, resulting in hazardous levels of laser radiation exposure to the operator.

Electromagnetic Compatibility

The Paladin laser system complies with the European requirements for electromagnetic compatibility as defined in the Electromagnetic Compatibility Directive 2014/30/EU.

The Paladin laser system is intended for use in an Industrial Environment. Operation of this laser system in a different EMC environment (Residential, for example) may require that the user take remedial action in addition to the normal installation, operation, and maintenance described in this manual to resolve potential electromagnetic compatibility problems. Coherent makes no claims beyond those listed below concerning the compatibility of this laser system in EMC environments other than the Industrial environment.

Coherent Inc. declares that the Paladin laser system meets the requirements of the EMC Directive 2014/30/EU based on the following standards:

- CISPR 11:2009 + A1:2010
- EN 55011, class A
- IEC 61000-3-2, -3
- IEC 61326-1:2012, Susceptibility
- IEC 61000-4-2, 3, 4, 5, 6, 8, 11

Environmental Compliance

RoHS Compliance

The RoHS directive restricts the use of certain hazardous substances in electrical and electronic equipment. Coherent can provide RoHS certification upon request for products requiring adherence to the RoHS Directive.

Compliance of this laser with the EMC requirements is certified by the CE mark.

China-RoHS Compliance

The China-RoHS directive restricts the use of certain hazardous substances in electrical and electronic equipment and applies to the production, sale, and import of products in the Peoples Republic of China. Refer to the table below for product components that are China-RoHS compliant.

Table 1-1. China-RoHS Compliance

铅	汞	镉	六价铬	多溴联苯	多溴二苯醚
Pb	Hg	Cd	$Cr6^{+}$	PBB	PBDE
X	O	O	O	O	O

O≡小于最高浓度值 X≡大于最高浓度值

Waste Electrical and Electronic Equipment (WEEE)

The European Waste Electrical and Electronic Equipment (WEEE) Directive (2012/19/EU) is represented by a crossed-out garbage container symbol (Figure 1-2 items 4 and 9). The purpose of this directive is to minimize the disposal of WEEE as unsorted municipal waste and to facilitate its separate collection.

Maximum Noise Level dB(A)

The noise level of the Paladin Laser System is under 80 dB(A). No sonic hazard is present.

Label Location and Description

Refer to the following figures and tables in this chapter for locations and descriptions of all labels plaques and engravings. The power supply and laser head is shipped with all labels placed as shown to meet safety, certification and identification requirements. These include warning labels indicating removable or replaceable protective housings, apertures through which laser radiation is emitted and labels of certification and identification [CFR 1040.10(g), CFR 1040.2, and CFR 1010.3/ EN 60825-1, Clause 5].

NOTICE
Labels must be applied at installation to meet safety, certification and identification requirements.

Laser Head

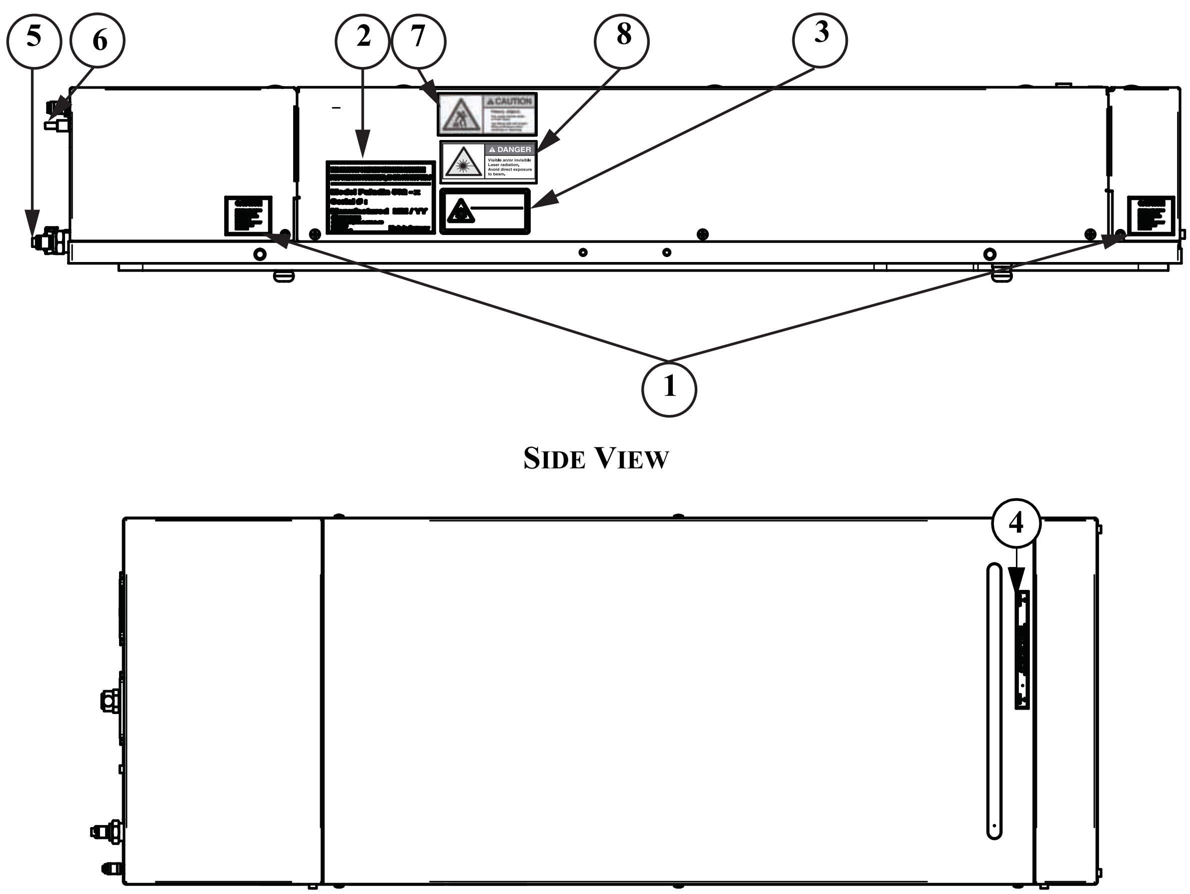


Figure 1-1. Label Locations- Laser Head

Table 1-2. Label Descriptions- Laser Head

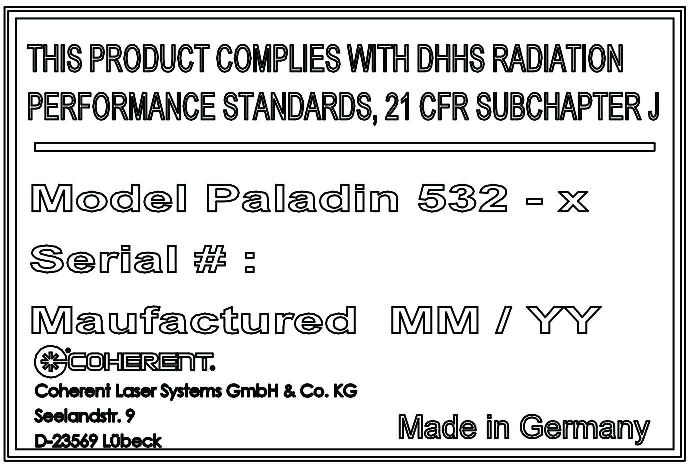
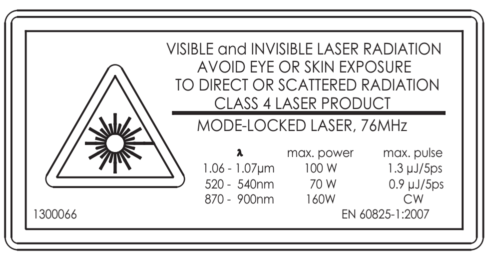
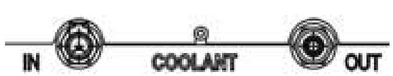
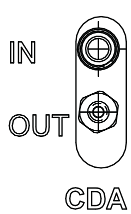
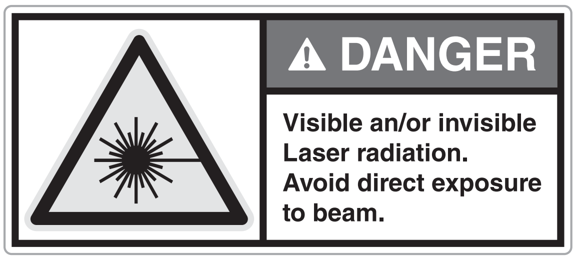
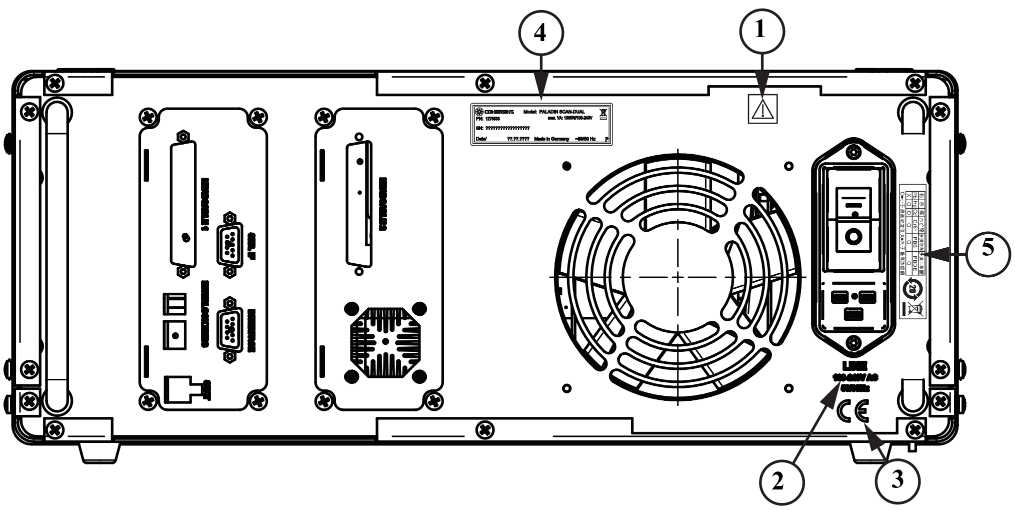
ITEM	LABEL	DESCRIPTION
1		**Caution Label:** Cautions the user of possible exposure to laser radiation if the cover is removed.
2		**Product Identification Label:** This label states the compliance to DHHS Performance Radiation Standards 21 CFR Ch. I, EN 60825-1 for this product. It also contains the model, serial number, manufacturing date, part number and product origin. Contains the European Waste Electrical and Electronic Equipment (WEEE) Directive Label (lower right corner): See "Waste Electrical and Electronic Equipment (WEEE, 2002)" on page 1-6.
3		**Laser Product Label:** This label describes the specific wavelength and output power level capabilities of the laser head. It also includes the laser class of the product. For more information, refer to "Laser Classification" on page 1-6
4		**Avoid Exposure Label:** This label identifies the location of the output beam.
5		**Coolant In/Out:** This engraving shows the location for the coolant inlet/outlet connection.
6		**CDA In/Out:** This engraving shows the location for the CDA inlet/outlet connection.

Table 1-2. Label Descriptions- Laser Head (Continued)

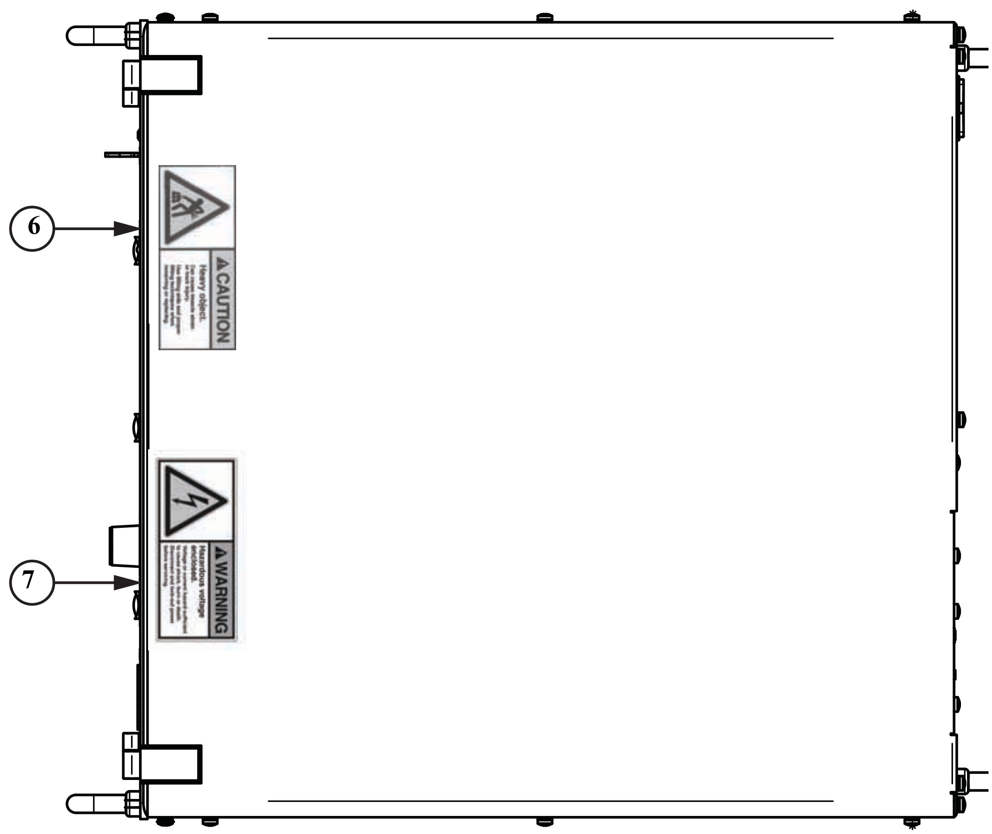
ITEM	LABEL	DESCRIPTION
7		**Heavy Object Label:** This label indicates that proper techniques and aids must be used when lifting the object.
8		**Laser Radiation Danger Label:** This label indicates the danger of visible and/or invisible.

Power Supply



REAR VIEW

Figure 1-2. Label Locations- Power Supply



TOP DOWN VIEW

Figure 1-2. Label Locations- Power Supply (Continued)

Table 1-3. Label Descriptions- Power Supply

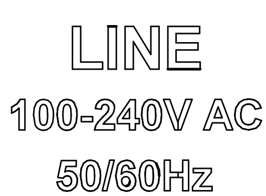
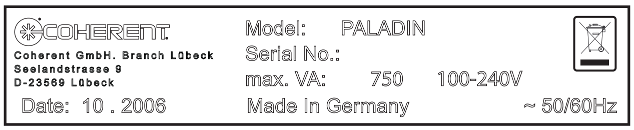
ITEM	LABEL	DESCRIPTION
1		**Important Information Label:** Identifies the presence of important operating and maintenance instructions. Use caution when working near this area of the system.
2		**Line Voltage Engraving:** Identifies line voltage specifications.

Table 1-3. Label Descriptions- Power Supply (Continued)

ITEM	LABEL	DESCRIPTION
3		**CE Certification Engraving:** Complies with the low voltage directive. Refer to “Safety Features and Compliance to Government Regulations” on page 1-6.
4		**Power Supply Identification Label:** Contains the manufacturing date, model number, serial number and product origin. Also contains the WEEE symbol that is described in “Waste Electrical and Electronic Equipment (WEEE)” on page 1-10.
5		**China RoHS Label:** Standardized label which meets substance labeling requirements for China RoHS. States 20 year Environment-Friendly Use Period (EFUP) warning for Lead (Pb) only. Also contains the WEEE symbol that is described on page 1-6.
6		**Heavy Object Label:** This label indicates that proper techniques and aids must be used when lifting the object.
7		**Hazardous Voltage Danger Label:** This label indicates the danger of hazardous voltage. Proper Lock-Out techniques must be used before service of the system.

French Translation/ Traduction Française Section 1: Sécureté

Introduction

Ce système Paladin Scan Laser a été conçu pour assurer la protection de l'opérateur en cas de panne quelconque de toute pièce isolée, pourvu que le système soit installé et utilisé correctement et de la manière décrite dans ce Manuel d'utilisation. Tous les utilisateurs du laser doivent lire, comprendre et suivre les avertissements sur la sécurité et les instructions d'utilisation contenus dans ce Manuel d'utilisation.

Étant donné que le fabricant n'est pas en mesure de garantir la sécurité des utilisateurs du laser en cas de panne de deux composants indépendants, il est impératif de ne pas faire fonctionner cet équipement si l'on découvre une mise en danger quelconque du personnel, une panne de composant, une installation incorrecte ou des dommages significatifs. La panne d'un composant unique n'est pas dangereuse, mais elle risque d'empêcher de protéger les utilisateurs ou d'empêcher de les avertir d'une condition dangereuse si une autre panne se produisait. Inspectez donc régulièrement le système laser pour y découvrir des dangers potentiels ou des pannes de composant qui compromettent la sécurité.

Si l'on suspecte que le système laser manque de pièces associées à la sécurité, qu'il est endommagé ou qu'il présente tout autre danger, le mettre hors tension, le débrancher immédiatement de la puissance d'entrée et contacter le représentant de service local de Coherent. N'utilisez le laser qu'une fois que tous les dangers potentiels qui compromettent sa sécurité ont été éliminés. Ne retirez aucun boîtier ou couvercle de protection, à moins d'en avoir reçu l'ordre de la part d'un représentant de service certifié par Coherent.

Ces informations destinées aux utilisateurs sont conformes aux nor-mes suivantes concernant les produits électroluminescents IEC 60825-1 / EN 60825-1 *« Sécurité des produits laser (Safety of laser products)—Partie 1 : Classification des équipements et exigences (Equipment classification and requirements) »* 21 Titre CFR 21 Chapitre 1, Sous-chapitre J, Partie 1040 *« Normes de performance pour les produits électroluminescents (Performance standards for light-emitting products) »*.

AVERTISSEMENT !
RAYONNEMENT LASER - ÉVITER L'EXPOSITION DE LA PEAU ET DES YEUX AU RAYONNEMENT DIFFUSÉ OU DIRECT - PRODUIT LASER DE CLASSE 4 !

AVERTISSEMENT !
L'utilisation de commandes, de réglages ou la réalisation de procédures autres que celles indiquées dans ce manuel peut entraîner une exposition à des rayon-nements dangereux.

Cette section relative à la sécurité des lasers doit être lue attentivement avant de faire fon-ctionner le système Paladin. Les instructions concernant la sécurité présentées dans ce manuel doivent être soigneusement suivies.

Risques

Les risques associés aux lasers concernent d'une manière générale les caté-gories suivantes :

- Risques biologiques dus à une exposition à des rayonnements laser pouvant endommager les yeux ou la peau
- Risques électriques liés à l'alimentation en électricité du laser ou aux circuits électriques asso-ciés
- Risques chimiques liés à la mise en contact du faisceau laser avec des substances volatiles ou inflammables, ou liés à des substances dégagées lors du traitement au laser de certaines matières

La liste présentée ci-avant ne prétend pas être exhaustive. Toute personne utilisant le laser doit prendre en compte l'interaction de l'appareil laser avec son environnement de travail particulier, afin d'identifier les risques éventuels.

Sécurité oculaire

La lumière laser, en raison de ses caractéristiques particulières, pose des risques en matière de sécurité que l'on ne retrouve pas dans les sources de lumière traditionnelles. Une utilisation sans danger de lasers n'est possible qu'à condition que tous les opérateurs du laser, ainsi que toutes personnes se trouvant à proximité du système laser, soient pleinement conscients des dangers encourus. Une utilisation sans danger de ce laser dépend du degré de familiarité de l'utilisateur avec l'instrument et des propriétés des faisceaux intenses de rayonnement cohérent de lumière.

Les précautions en matière de sécurité énumérées ci-après doivent être lues et respectées par toute personne travaillant avec ou à proximité du laser. Il convient de s'assurer qu'à tout moment, l'ensemble du personnel chargé du fonctionnement du laser, de son entretien ou de sa réparation est protégé de toute exposition, accidentelle ou inutile, à des rayonnements laser dépassant les limites d'émission acceptées, définies dans les normes de sécurité des lasers.

AVERTISSEMENT !
Un contact oculaire direct avec le faisceau de sortie peut entraîner des lésions oculaires graves, pouvant aller jusqu'à la cécité.

La sécurité oculaire doit être la préoccupation numéro une lors de l'utilisation d'un laser. En plus du faisceau principal, il existe souvent des sous-faisceaux de taille moins importante à proximité du système laser, présents sous divers angles. Ces faisceaux se forment par un phénomène de réflexion spéculaire du faisceau principal sur des surfaces brillantes, telles que des lentilles ou des séparateurs de faisceaux. Bien que plus faible que celle du faisceau principal, l'intensité de ces faisceaux peut encore être suffisante pour entraîner des lésions oculaires.

Les rayons laser sont suffisamment puissants pour brûler la peau, les vêtements ou les matériaux comb-ustibles, même à une certaine distance. Ils peuvent faire s'enflammer des substances volatiles, telles que l'alcool, l'essence, l'éther et autres solvants, et peuvent endommager les éléments photosensibles des caméras vidéo, des photomu-ltiplicateurs et des photo-diodes. Il est conseillé à l'utilisateur de respecter les mesures de contrôle répertoriées ci-dessous.

Précautions et recommandations

1. Respectez toutes les précautions de sécurité données dans les ma-nuels de pré-installation et d'utilisation.
2. Toujours porter une protection oculaire appropriée contre les longueurs d'onde spécifiques et l'énergie laser générées par l'appareil.
3. Toujours protéger la peau contre les longueurs d'onde spécifiques et l'énergie laser générées par l'appareil.
4. Évitez de porter des montres, des bijoux ou tout objet susceptible de réfléchir ou de diffuser le rayon laser.

5. Restez conscient de la trajectoire du faisceau laser, tout particulièrement lorsque des dispositifs optiques externes sont utilisés pour diriger le rayon.
6. Procurez un périmètre de protection contre les faisceaux laser, chaque fois que cela est possible.
7. Utilisez des cibles d'absorption d'énergie adaptées pour bloquer le faisceau.
8. Bloquez le faisceau avant d'utiliser des outils, tels que des clés hexagonales ou des clés Balldriver® sur les dispositifs optiques externes.
9. Restreignez l'accès au laser à des utilisateurs formés et qualifiés, connaissant bien les précautions de sécurité à observer en présence d'un laser. Lorsqu'ils ne sont pas utilisés, les lasers doivent être éteints complètement et leur accès doit être interdit au person-nel non autorisé.
10. Obstruez le rayon laser à l'aide d'un matériau absorbant la lumière. Un faisceau de lumière laser peut rester collimaté sur de longues distances, ce qui pose un risque potentiel dans le cas où il n'est pas confiné. Il est recommandé de faire fonctionner le laser dans une pièce fermée.
11. Placez des panneaux d'avertissement sur le laser dans la zone du rayon laser, afin d'alerter les personnes présentes.
12. En cas de manipulation de solvants dans la zone du laser, prenez des précautions extrêmes.
13. Ne regardez jamais directement la source de lumière laser, ou la lumière laser diffusée par une surface réfléchissante, même si vous portez des lunettes de protection contre le laser. Ne jamais fixer le faisceau laser en face.
14. Réglez le laser de manière à ce que la hauteur du faisceau soit largement en dessous ou largement au-dessus du niveau des yeux.
15. Évitez toute exposition directe à la lumière du laser. Les rayons laser peuvent très facilement brûler la peau, ou mettre feu aux vêtements. Portez des vêtements qui conviennent à la longueur d'onde générée afin de protéger votre peau contre le rayonnement laser.
16. Faites part de toutes ces précau-tions à toutes les personnes travaillant avec ou à proximité du laser.

ATTENTION !
Les lunettes de protection contre le laser protègent l'utilisateur des expositions accidentelles aux rayonnements laser, en bloquant les longueurs d'onde laser de la lumière. Cependant, les lunettes de protection contre le laser peuvent également empêcher l'opérateur de voir le faisceau ou le faisceau ponctuel. Faites preuve d'une extrême prudence, même lorsque vous portez des lunettes de protection.

Sécurité électrique

Ce système Paladin Scan Laser contient des tensions potentiellement dangereuses à l'intérieur des boîtiers de protection de l'alimentation et de la tête du laser. Ne pas ouvrir ces boîtiers de protection. Ne pas faire fonctionner le système laser sans borne de protection (sécurité) de mise à la terre branchée au raccord d'admission d'alimentation du site.

Il est nécessaire d'effectuer une évaluation générale des risques et de l'application avant de faire fonctionner cette dernière.

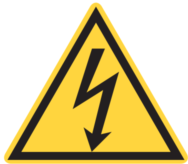
Cet équipement ne doit être utilisé qu'en intérieur (salle propre climatisée seulement) dans des emplacements sans danger ; seul un personnel qualifié et formé sur son utilisation doit le faire fonctionner.

Le câble d'alimentation principal doit être conforme aux normes nationales et/ou aux codes électriques du pays dans lequel le système est utilisé. Il ne faut utiliser que des câbles d'alimentation principale mis à la terre.

Le câble de raccord à l'alimentation principale doit avoir la tension nominale correspondant au courant maximal de l'équipement et ne doit pas mesurer plus de 3 mètres.

Le dispositif de déconnexion doit se trouver électriquement aussi près que possible du niveau pratique de la source d'alimentation. Il ne faut pas placer électriquement des pièces qui consomment de l'alimentation entre la source d'alimentation et le dispositif de déconnexion ; néanmoins, les circuits de suppression de l'interférence électromagnétique sont autorisés du côté de l'alimentation du dispositif de déconnexion.

Des dispositifs de protection contre une surtension seront fournis au moment de l'installation finale de la manière requise par les instructions d'installation.



DANGER !
Le fonctionnement normal du Paladin ne nécessite pas d'accès aux circuits de l'alimentation électrique. Le fait de retirer le couvercle protecteur de l'alimentation expose l'utilisateur à des risques électriques potentiellement mortels. Faites appel à un représentant du service autorisé avant d'essayer de corriger un problème lié à l'alimentation électrique.

Précautions et recommandations

Les précautions suivantes doivent être observées par toute personne travaillant avec des circuits électriques potentiellement dangereux :

DANGER !
Les règles en matière de sécurité électrique doivent être stricte--ment respectées lorsque vous travaillez avec des systèmes d'alimentation électrique. Dans le cas contraire, vous serez exposés à des niveaux électriques mortels.

1. Débranchez l'alimentation secteur avant de travailler sur un équipement électrique lorsqu'il n'est pas nécessaire pour vous que l'équipement soit en train de fonctionner.
2. La sortie d'alimentation électrique ne doit pas être court-circuitée ou mise à la terre. Une protection efficace contre des risques électriques potentiels nécessite que le câble d'alimentation soit correctement relié à la borne de terre, et nécessite une mise à la terre externe adaptée. Vérifiez ces raccordements au moment de l'installation, et revérifiez-les de manière régulière par la suite.
3. Ne travaillez jamais avec des équipements électriques en l'absence d'une autre personne connaissant bien le fonctionnement de l'équipement, ainsi que les risques associés à ce fonctionnement et compétente pour administrer les premiers soins.
4. Lorsque c'est possible, ne touchez l'équipement qu'avec une seule main pour éviter le risque qu'un courant électrique ne circule dans votre corps, dans le cas où vous seriez accidentellement en contact avec un circuit alimenté.
5. Utilisez toujours des outils homologués, isolés électriquement.
6. Des techniques de mesure spéciales sont nécessaires pour ce système. Seul un technicien disposant d'une compréhension

complète du fonctionnement du système et des circuits électroniques associés peut choisir les référen-ces au sol.

Dispositifs de sécurité et Conformité aux réglementations gouvernementales

Les dispositifs suivants ont été intégrés au Paladin système laser afin de se conformer à plusieurs exigences gouvernementales. Les exigences gouvernementales américaines applicables sont répertoriées dans le 21 CFR, sous-chapitre J, partie II, administré par le Centre américain des appareils et de la santé radiologique (Center for Devices and Radiological Health, CDRH). Les exigences de la Communauté européenne en matière de sécurité des produits sont spécifiées dans la directive basse tension (Low Voltage Directive, LVD) (2014/35/EU). Cette directive basse tension exige que les lasers soient conformes aux normes UL 61010-1:2012 et CAN/CSA-C22.2 No. 61010-1-12, et EN60825-1 « Radiation Safety of Laser Products » (Sécurité du rayonnement des produits laser). La conformité de ce laser aux exigences (LVD) est certifiée par la marque CE.

Classification laser

Les normes et exigences gouvernementales précisent que le laser doit être classé en fonction de sa puissance de sortie ou de son énergie et de sa longueur d’onde. Le Paladin est classé dans la Catégorie IV selon le code 21 CFR, sous-chapitre J, partie II, section 1040-10 (d). Dans le cadre des normes de la Communauté européenne, ce laser Paladin est classé dans la Catégorie 4 selon la réglementation EN 60825-1, clause 9. Dans le présent manuel, la classification de référence sera la Classe 4.

Enveloppe protectrice

La tête du laser est renfermée dans un boîtier de protection qui empêche aux personnes d’entrer en contact avec un rayonnement qui dépasse les limites de rayonnement autorisées pour la Catégorie, de la manière spécifiée dans le code 21CFR, Partie 1040, Section 1040.10 (f)(1) et la norme EN 60825-1/IEC 60825-1 Clause 6.2, sauf en ce qui concerne le faisceau de sortie, qui est de Catégorie 4.

Verrouillage de l’obturateur

Le verrouillage de l'obturateur se trouve à l'arrière de l'alimentation (voir «EMBS» dans Figure 4-2). L’obturateur ne s’ouvrira pas à moins que le circuit de verrouillage ne soit fermé. Si le circuit du verrouillage de l’obturateur (EMBS) est ouvert pendant que le laser fonctionne, il entraînera la fermeture de l’obturateur sans mettre le laser hors tension.

Pour l'EMBS, un court-circuit entre les deux tiges du haut permet à l'obturateur de fonctionner normalement, tandis qu'un circuit ouvert empêche à l'obturateur d'être ouvert. L'impédance du court-circuit doit être inférieure à 100 Ohms. L'impéd-ance du circuit ouvert doit être supérieure à 5 K Ohms, en donnant préférence à une isolation galvanique. Le courant à travers un court-circuit sera typiquement de 20 mA.

Opération de l'obturateur

L'obturateur de faisceau électro-magnétique (Electro-Magnetic Beam Shutter, EMBS) est un dispositif de sécurité qui a trait à la norme EN ISO 13849-1 catégorie 2. Il est normalement fermé et se ferme dans le sens de la gravité lorsque la tête est placée avec ses pieds en direction du sol. Le statut de l'obturateur est continuellement surveillé par le biais de deux barrières lumineuses, une pour le statut Arrêt et l'autre pour le statut En marche.

L'obturateur se ferme en mode de fonctionnement standard au bout de 10 ms. Le délai de réac-tion de l'obturateur en réponse à une alarme est de 25 ms.

AVIS
L'obturateur est un dispositif de sécurité obligatoire pour le fonctionnement du laser. Il ne doit pas être utilisé comme moyen de contrôle de la durée d'exposition !

Verrouillage externe

Le système ne fonctionne pas avec le verrouillage ouvert. Un connecteur EXTERNAL INTERLOCK est situé sur le panneau arrière du bloc d'alimentation (voir «Interlock» dans Figure 4-2). Le statut de verrouillage est surveillé par l'UC de la carte de contrôle située dans la source d'alimentation principale. Si le verrouillage est ouvert, un message apparaît sur le panneau avant de la source d'alimentation.

Pour intégrer correctement le laser au contrôle de sécurité de l'application, un circuit de verrouillage externe doit être connecté au système laser et câblé à un commutateur de porte ou panneau d'accès afin de procurer une sécurité de fonctionnement supplémentaire. Lorsqu'une porte est ouverte, le laser s'arrête et l'obturateur se ferme. La condition de faute doit être effacée et les diodes mises en marche pour redémarrer le laser.

Pour incorporer un circuit de verrouillage de sécurité externe au système laser, mettez le laser hors tension, puis retirez le cavalier du connecteur de verrouillage sur le panneau arrière de la source d'alimentation principale. Joignez à ce connecteur un circuit de verrouillage externe fourni par l'utilisateur. Tout circuit de ce type doit être équivalent à un dispositif de fermeture mécanique du circuit. Il ne faut en aucune circonstance connecter une tension ou une source de courant externe à ce circuit. La circuiterie du verrouillage externe doit être isolée de tout autre circuit ou mise à la terre électrique.

Une interruption de ce circuit arrête les diodes du laser et élimine l'émission de lumière haute dangereuse. La boucle de verrouillage est -établie en tant que boucle de courant de 20 mA, qui doit être fermée pour activer le rayonnement laser.

L'intégrateur doit assurer le câblage correct du système. Cela comprend la protection du câblage au niveau de l'alimentation électrique, de la boucle de verrouillage et de l'EMBS contre un court-circuit ou un court-circuit transversal. L'intégrateur doit éviter toute alimentation aux entrées d'asservissement (boucle de verrouillage, EMBS) du système.

Les entrées du « verrouillage de l'obturateur » et du « verrouillage de sécurité du laser » de la source d'alimentation doivent être connectées au circuit d'ARRÊT d'URGENCE (EMERGENCY STOP) et/ou au circuit de SUSPENSION d'URGENCE (EMERGENCY OFF) de l'application finale de la manière exigée lors de l'évaluation des risques de l'application finale.

Indicateurs d'émission laser

Les voyants adéquatement étiquetés sur les deux sources d'alimentation et la tête du laser s'allument lorsque le laser sur le bouton est configuré sur la position LASER EN MARCHE (LASER ON) avant que l'émission laser ne puisse se produire. Des voyants blancs ou jaunes sont utilisés et sont visibles lorsqu'on porte des lunettes de sécurité.

Boutons de commande

Les boutons de commande du laser sont placés de telle sorte que l'opérateur ne soit pas exposé à des émissions laser lorsqu'il manipule ces boutons.

DANGER !
Utiliser les boutons de commande d'une autre manière que celle décrite dans ce manuel risque de compromettre la protection fournie par le système et d'entraîner des niveaux dangereux d'exposition au rayonnement laser pour l'opérateur.

Compatibilité électromag-nétique

Le système Paladinlaser est conforme aux exigences européennes concernant la compatibilité électromagnétique, telles qu'elles sont définies dans la directive sur la compatibilité électromagnétique 2014/30/EU.

Ce système laser Paladin doit être utilisé dans un environ-nement industriel. L'exploitation de ce système laser dans un environ-nement EMC différent, par exemple, résidentiel, pourra demander que l'util-isateur prenne des mesures correc-tives en plus de l'installation, de l'opération et de la maintenance normales décrites dans ce manuel, ceci afin de résoudre les problèmes potentiels de compatibilité élec-tro-magnétique. Coherent ne donne aucune garantie au-delà de celles répertoriées ci-dessous concernant la compatibilité de ce système laser dans des environnements EMC autres qu'un environnement industriel.

Coherent Inc. déclare que ce système laser Paladin répond aux exigences de la directive EMC 2014/30/EU en se fondant sur les normes suivantes :

- CISPR 11:2009 + A1:2010
- EN 55011, catégorie A
- IEC 61000-3-2, -3
- IEC 61326-1:2012, Susceptibilité
- IEC 61000-4-2, 3, 4, 5, 6, 8, 11

Respect de l'environnement

Conformité RoHS

La directive RoHS restreint l'utilisation de certaines substances dangereuses dans les équipements électriques et électroniques. Coherent peut fournir une certification RoHS sur demande, pour les produits ayant pour obligation de respecter la directive RoHS.

La conformité de ce laser avec les exigences EMC est certifiée par la marque CE.

Conformité China-RoHS

La directive China-RoHS restreint l'utilisation de certaines substances dangereuses dans un équipement électrique et électronique et s'applique à la production, vente et importation de produits dans la République populaire de Chine. Consultez le tableau ci-dessous pour connaître la liste des composants du produit qui sont conformes à la directive China-RoHS.

Table 1-4. Conformité China-RoHS

铅	汞	镉	六价铬	多溴联苯	多溴二苯醚
Pb	Hg	Cd	$Cr6^+$	PBB	PBDE
X	O	O	O	O	O

O=小于最高浓度值 X=大于最高浓度值

Déchets électriques et électroniques (DEEE)

La directive européenne sur les déchets électriques et électroniques (DEEE) (2012/19/EU) est représentée par un logo représentant une poubelle barrée. L'objectif de cette directive est de minimiser l'élimination des déchets électriques et électroniques en tant que déchets municipaux non triés et de faciliter le ramassage sélectif.

Niveau de bruit maximal en dB(A)

Le niveau de bruit du système laser Paladin est inférieur à 80 dB(A). Aucun danger sonique n'existe.

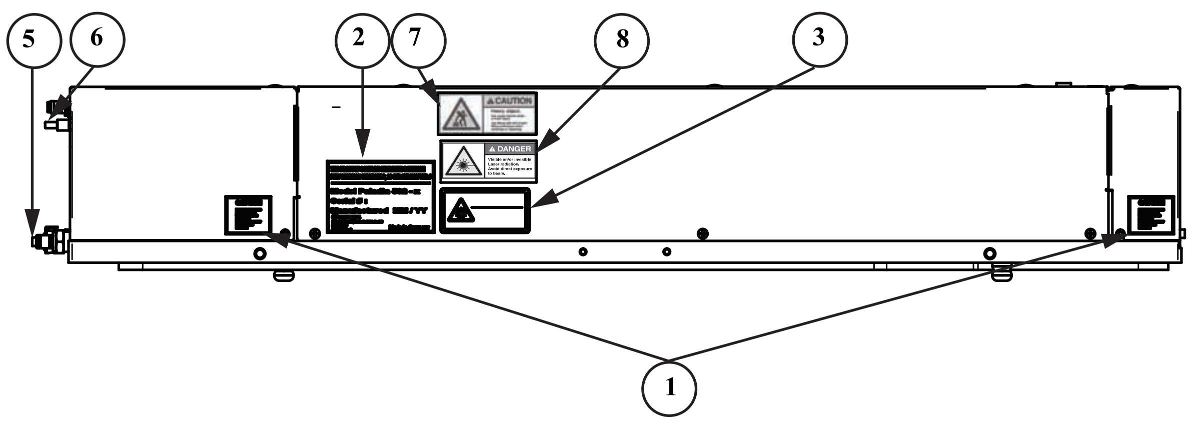
Emplacement et description des étiquettes

Consultez les figures et tableaux ci-après dans ce chapitre pour connaître les emplacements et descriptions de toutes les plaques et gravures d'étiquetage. La source d'alimentation et la tête du laser sont expédiées avec toutes les étiquettes placées de la manière indiquée pour répondre aux exigences de sécurité, certification et identification. Ces étiquettes comprennent des étiquettes d'avertissement indiquant les gaines protec-trices amovibles ou déplaçables, ainsi que les orifices à travers lesquels des rayonnements laser sont émis, et également des étiquettes de certification et d'identification [CFR 1040.10(g), CFR 1040.2 et CFR 1010.3/ EN60825-1, Clause 5].

AVIS

Les étiquettes doivent être appliquées à l'équipement au moment de l'installation pour répondre aux exigences de sécurité, certifica-tion et identification.

Tête laser



VUE LATÉRALE

Figure 1-3. Emplacements des étiquettes - Tête du laser

VUE DU HAUT VERS LE BAS

Figure 1-3. Emplacements des étiquettes - Tête du laser (Continued)

Table 1-5. Définitions des étiquettes - Tête du Laser

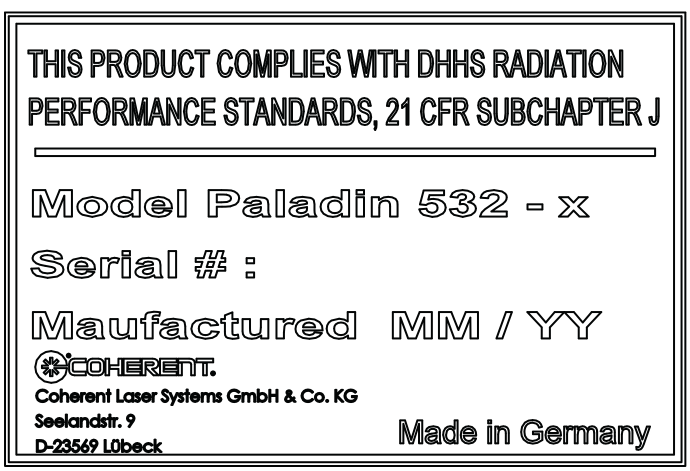
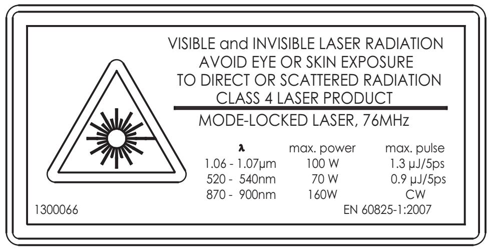
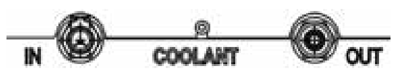
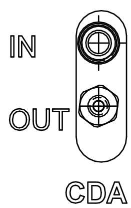
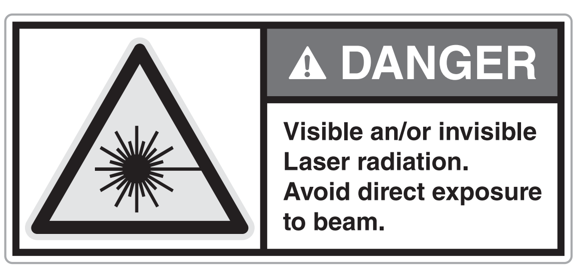
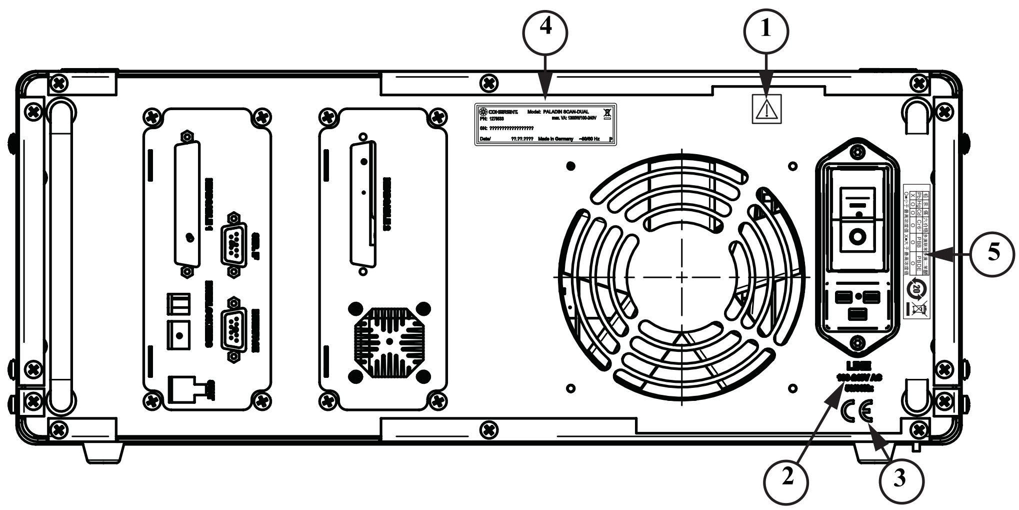
ÉLEMENT	ÉTIQUETTE	DESCRIPTION
1		**Étiquette « Caution » (Mise en garde) :** elle prévient l'utilisateur de la possibilité d'une exposition au rayonnement laser si le couvercle est retiré.
2		**Étiquette d'identification du produit :** cette étiquette énonce la conformité de ce produit aux Normes de performance sur le rayonnement de la DHHS 21 CFR Ch. I, EN 60825-1. Elle porte aussi le numéro du modèle, le numéro de série, la date de fabrication, le numéro de pièce et l'origine du produit. Elle porte l'étiquette de la directive européenne sur les Déchets élec-triques et électroniques (DEEE) (coin inférieur droit) : voir Déchets électriques et électroniques (DEEE, 2002) à la page 1-6.

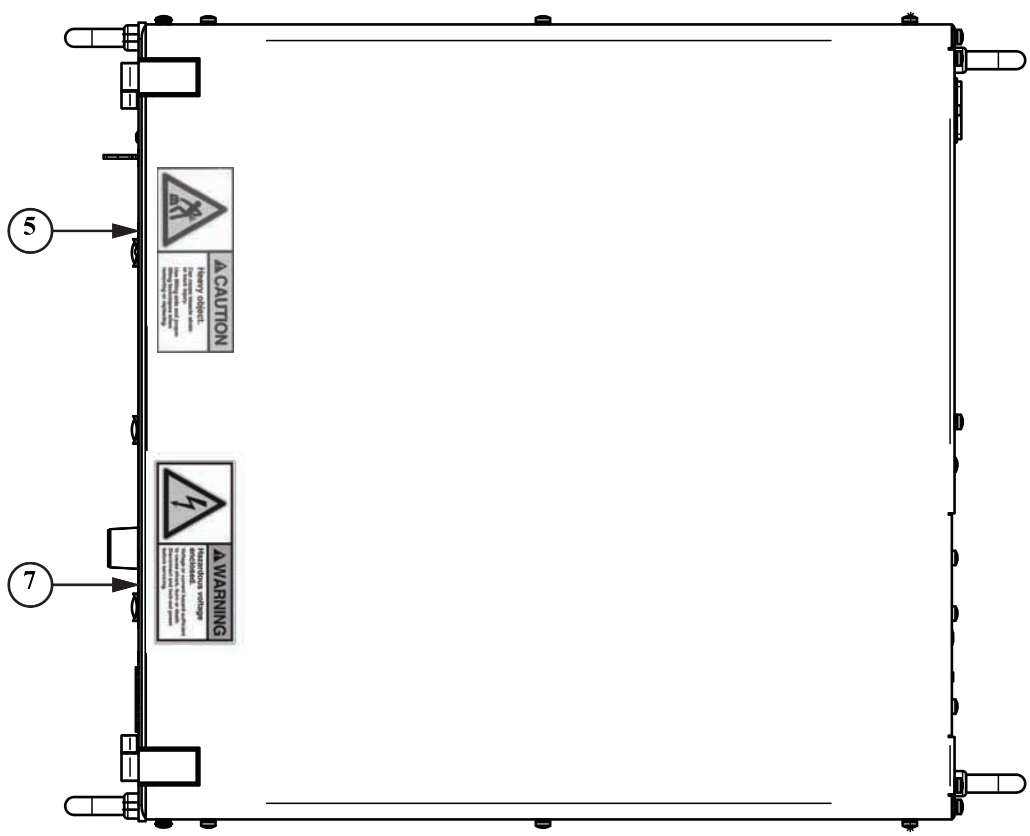
Table 1-5. Définitions des étiquettes - Tête du Laser

ÉLEMENT	ÉTIQUETTE	DESCRIPTION
3		**Étiquette de produit laser :** cette étiquette décrit la longueur d'onde spécifique et les capacités de niveau de puissance de sortie de la tête du laser. Elle porte aussi la catégorie de laser de ce produit. Pour un complément d'information, consultez "Laser Classification" on page 1-6
4		**Étiquette « Avoid Exposure » (Éviter toute exposition) :** cette étiquette identifie l'emplacement du faisceau de sortie.
5		**« Coolant In/Out » (Entrée/Sortie du liquide de refroidissement) :** cette plaque gravée montre l'emplacement du raccord d'entrée/sortie du liquide de refroidissement.
6		**« CDA In/Out » (Entrée/Sortie ASP) :** cette plaque gravée montre l'emplacement du raccord d'entrée/sortie de l'ASP (air sec propre).
7		**Étiquette « Heavy Object » (Objet lourd) :** cette étiquette indique qu'il faut utiliser des techniques et aides adéquates lorsqu'on soulève cet objet.
8		**Étiquette de danger de rayonnement laser :** cette étiquette indique le danger de rayonnement visible et/ou invisible.

Alimentation électrique du laser



VUE ARRIÈRE



VUE DU HAUT VERS LE BAS

Figure 1-4. Emplacements des étiquettes - Source d'alimentation

Table 1-6. Description des étiquettes - Source d'alimentation

ÉLEMENT	ÉTIQUETTE	DESCRIPTION
1		**Étiquette d'information importante :** elle identifie la présence d'instructions de fonctionnement et de maintenance importantes. Prendre ses précau-tions lorsqu'on travaille près de cette zone du système.
2	LINE 100-240V AC 50/60Hz	**Plaque gravée de tension de ligne :** elle identifie les spécifica-tions de tension de la ligne.
3	CE	**Plaque gravée de certification CE :** elle se conforme à la directive de basse ten-sion. Consultez "Safety Features and Compliance to Government Regulations" on page 1-6.
4	COHERENT Coherent GmbH. Branch Lübeck Seelandstrasse 9 D-23569 Lübeck Date: 10 . 2006 Model: PALADIN Serial No.: max. VA: 750 100-240V Made In Germany ~ 50/60Hz	**Étiquette d'identification de la source d'alimentation :** elle contient la date de fabrication, le numéro de modèle, le numéro de série et l'origine du produit. Elle contient aussi le symbole DEEE décrit dans "Déchets électriques et électroniques (DEEE)" on page 1-26

Table 1-6. Description des étiquettes - Source d'alimentation (Continued)

ÉLEMENT	ÉTIQUETTE	DESCRIPTION
5		**Étiquette China RoHS :** étiquette standardisée qui répond aux exigences d'étiquetage de substances conformément à la directive relative à la limitation de certaines substances dangereuses dans les équipements électriques et électroniques en Chine (Restriction of the Use of Certain Hazardous Substances, China RoHS). Elle donne un avertissement de période d'utilisation sans risques pour l'environnement (Environment-Friendly Use Period - EFUP) de 20 ans pour le plomb (Pb) seulement. Elle contient aussi le symbole DEEE décrit à la page 1- 26.
6		**Étiquette « Heavy Object » (Objet lourd) :** cette étiquette indique qu'il faut utiliser des techniques et aides adéquates lorsqu'on soulève cet objet.
7		**Étiquette de risque de tension dangereuse :** cette étiquette indique le risque d'une tension dangereuse. Des techniques de verrouillage adéquates doivent être utilisées avant de procéder à l'entretien de ce système.

SECTION TWO: DESCRIPTION AND SPECIFICATIONS

System Description

The Paladin Scan Laser is a compact diode-pumped solid-state modelocked laser that provides green (532 nm)output with pulse repetition rates in the 76 MHz region. Specifications for the Paladin are in Table 2-2.

The Paladin Scan Laser consists of the laser head and one power supply connected by two electrical cables. One diode module is located in the laser head.

Laser Head

The laser head contains the optical elements, a circuit board and a shutter. The major optical elements in the hermetically sealed head include a proprietary crystal set as the gain medium, a crystal as a second harmonic generation (SHG). All optical components are mounted on a kinematically mounted resonator for strength and stability.

The temperature of the gain medium is controlled by a thermo-electric cooler (TEC) that controls the temperature of the optical element. The temperature of the SHG crystal is set to approximately 150 °C and kept stable by an oven. Accumulated heat in the laser head is dissipated through the water cooled baseplate of the housing. The CPU in the power supply monitors the baseplate temperature. If the laser head temperature is too high, the system deactivates.

The laser head gain medium is used in an end pumped geometry, with the pump power provided by a fiber-delivered diode module. The distal end of the optical fiber is imaged into the gain medium.

Power Supply

The power supply provides all required use interfaces. It contains the primary power supply which generates all system voltages, the main system control board and connections for RS-232 communication, safety interlock and shutter control. Laser control can be facilitated by RS-232 communication or through the power supply front panel control buttons.

Laser Diode Assembly

The laser diode is located in the laser head and is hermetically sealed. The diode includes a circuit board with an EEPROM and a heat sink sensor.

Diode/Heat Sink Temperature

The laser diode temperatures are controlled passively by the water cooling provided for the laser head. Laser operation will stop if the laser diode temperature exceeds 35 °C.

Thermal Management of the Laser Head

Water Cooling

The heat generated by the laser diode and other components located inside the laser head must be dissipated by means of water cooling. Regulated water flow is used to transfer heat from the heatsink to the outside of the unit. To avoid build-up of condensation inside the laser diode module, water cooling must be turned off whenever the laser diode is turned off. See table 2-4 for cooling water requirements.

Specifications and Requirements

Dimensions and Weights

System dimensions and weights for the Paladin Scan Laser are given in Table 2-1. See "Dimensions" on page 2-6 for graphical representations of the system dimensions.

Table 2-1. Dimensions and Weights

DIMENSIONS	
Laser head (W x H x L)	350 x 165.2 x 930 mm (13.8 x 6.5 x 36.6 in)
Power supply	439 x 185 x536 mm (17.3 x 7.3 21.1 in)
Head Cable 1 Length	5 m (16.4 ft)
Head Cable 2 Length	5 m (16.4 ft)

Table 2-1. Dimensions and Weights (Continued)

WEIGHTS	
Laser head:	49.3 kg ± 5 % (108.7 lbs)
Power supply:	15.5 kg ± 5 % (34.2 lbs)

System Specifications

Specifications for the Paladin Scan Laser are provided in Table 2-2.

Table 2-2. Performance Specifications

PARAMETER	SPECIFICATION
Wavelength	532 nm
Output Power[a]	36 W
Repetition Rate	76.0 MHz ± 1 MHz
Pulse Width	< 23.3 ps FWHM
BEAM PROPERTIES	**SPECIFICATION**
M^2	<1.2
Nominal Waist Size	1.35 mm ± 0.135 mm
Waist Location X,Y	< ± 15% of Rayleigh Range
Beam Astigmatism	≤ 20% of Rayleigh Range @ t = 0; ± 15% over laser life (additional)
Centroid Position (laser to laser)	± 0.35 mm
Pointing Tolerance (laser to laser)	± 0.7 mrad
POLARIZATION	**SPECIFICATION**
State of Polarization	Linear, Vertical
Orientation of Linear Plane	± 3°
Power Ratio of Polarization (PY:PX)	> 200

a. Power not adjustable. The numbers given include tolerance ranges.

Utility Requirements

The utility requirements for the Paladin laser system are provided in Table 2-3.

Table 2-3. Utility Requirements

OPERATING VOLTAGE	
Power Supply:	100-240 VAC, 50/60 Hz
MAXIMUM POWER CONSUMPTION	
Power Supply:	1300 W
Maximum ambient temperature (operating): +30 °C (95 °F).	

Cooling Water Requirements

A chiller must be provided by the customer. The cooling water requirements for the Paladin Scan are given in Table 2-4.

Table 2-4. Cooling Fluid Requirements

DESCRIPTION	PARAMETER
Coolant Type	AQUA 90
Supply Temperature	18 °C (64 °F) (at laser supply)
Temperature Increase	2.0 K @ 18 °C (64 °F) at Ambient up to 2.15 K @ 30 °C (95 °F) at Ambient
Volume Flow[a]:	appr. 6 nlpm@18 °C (64 °F)
Pressure Drop:	ppr. 30 kPa@6 nlpm (4.4 psi)
Cooling Capacity	900 W @ Ambient up to 30 °C (95 °F)
Max. Pressure:	150 kPa (21.8 psi)
Filling Volume[b]:	0.6 L

a. To avoid build-up of condensation inside the laser diode module, water cooling must be turned off whenever the laser diode is turned off.
b. Volume is for the laser head only. Volume does not include hose and external chiller. Follow manufacturer instructions to fill the chiller correctly.

CDA Gas Requirements

Purge gas line must be provided by the customer. The purge gas requirements for the Paladin Scan are given in Table 2-5.

Table 2-5. Purge Gas Requirements

DESCRIPTION	PARAMETER
Pressure	>= 30 kPa (10.9 psi)
Temperature	20 °C ± 5 °C (68 °F ± 2.8 °F)
Volume Flow	> 6 nlpm (estimated; depending on specs below)
Quality[a]	particle filtration: 0.003 µm or better max. concentration TVOC: <0.2 µg/m3 max. concentration siloxanes: <0.1 µg/m3 max. concentration sulfates: <0.05 µg/m3 max. concentration anions/cations: <0.1 µg/m3 max. concentration ammonium: <0.1 µg/m3
Humidity	dew point below -90 °C (-130 °F) at standard atmospheric pressure (alarm level)

a. Any connection of the laser head to pneumatic systems with higher concentration of contaminants will cancel any claim of warranty!

Environmental Requirements

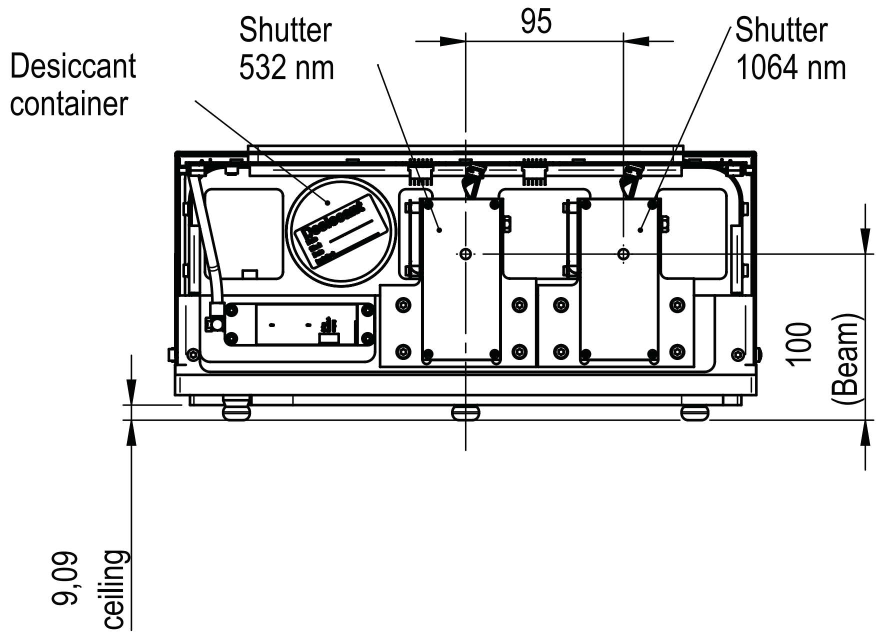
The Environmental requirements for the Paladin laser system are provided in Table 2-6.

Table 2-6. Environmental Requirements

DESCRIPTION	OPERATING	NON-OPERATING
Temperature:	18 °C to 30 °C (64 °F to 86 °F)	10 °C to 35 °C (50 °F to 95 °F)
Maximum Humidity: (non-condensing)	26 % to 95 %	≤ 70 %

Dimensions

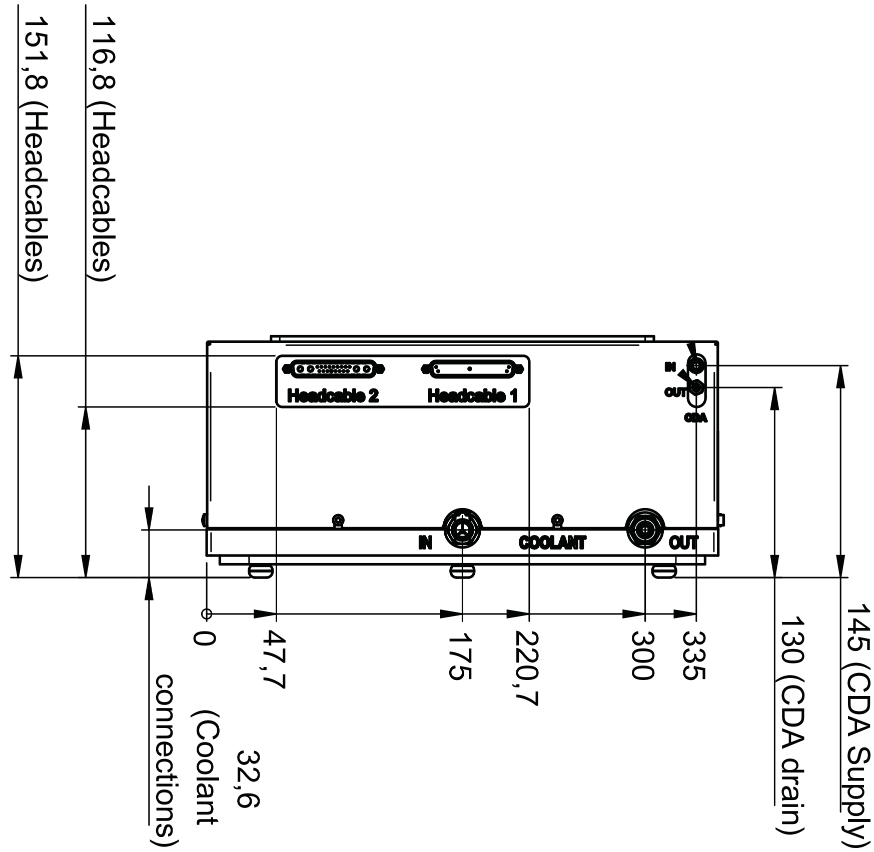
Laser Head



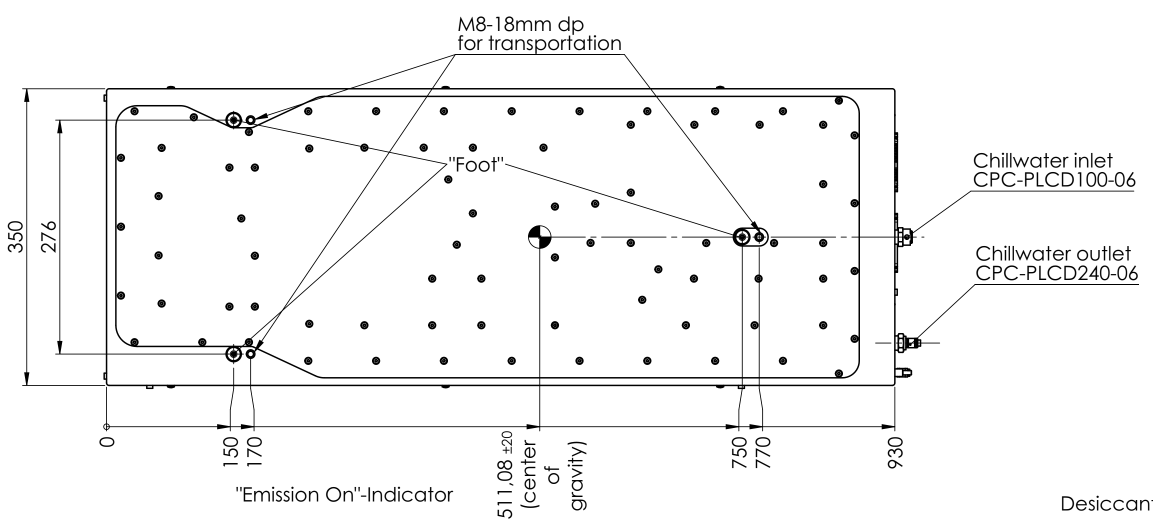
FRONT VIEW

Dimensions are given in (mm).

Figure 2-1. Paladin Scan Head Dimensions (Sheet 1 of 3)



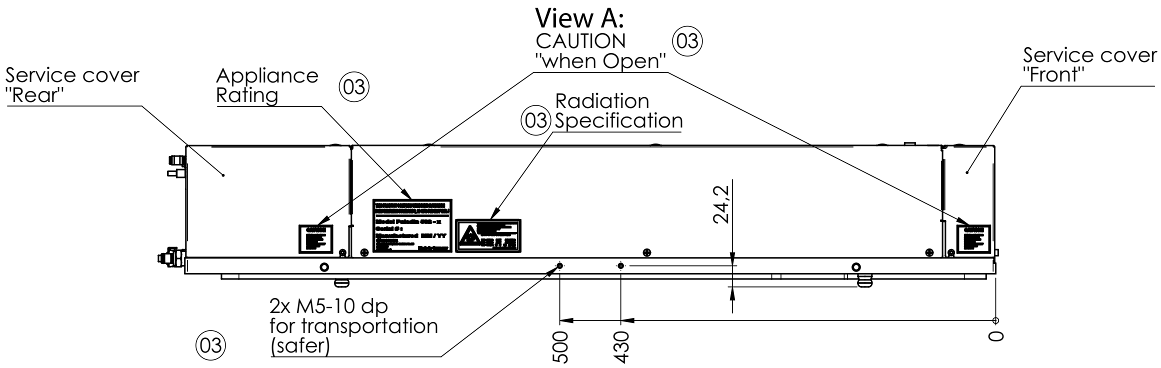
REAR VIEW

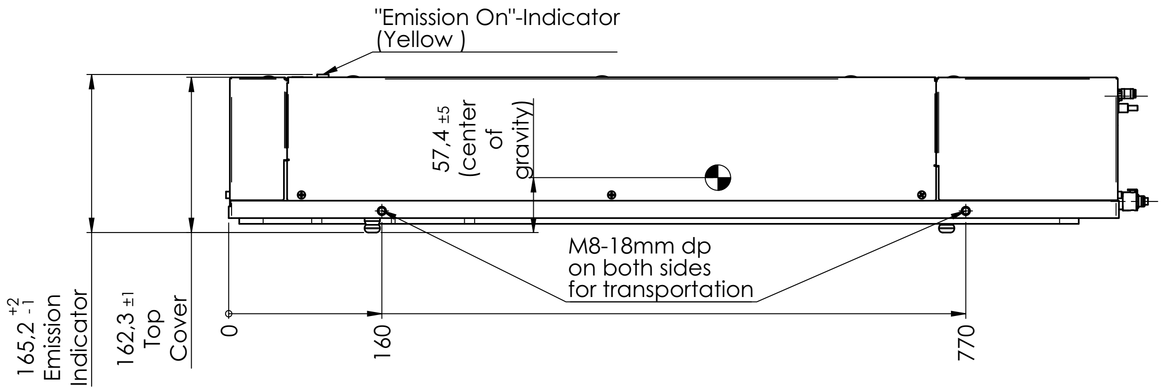


BOTTOM UP VIEW

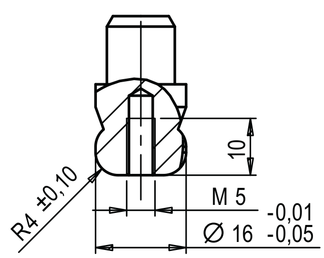
Dimensions are given in (mm).

Figure 2-1. Paladin Scan Head Dimensions (Sheet 2 of 3)





SIDE VIEW

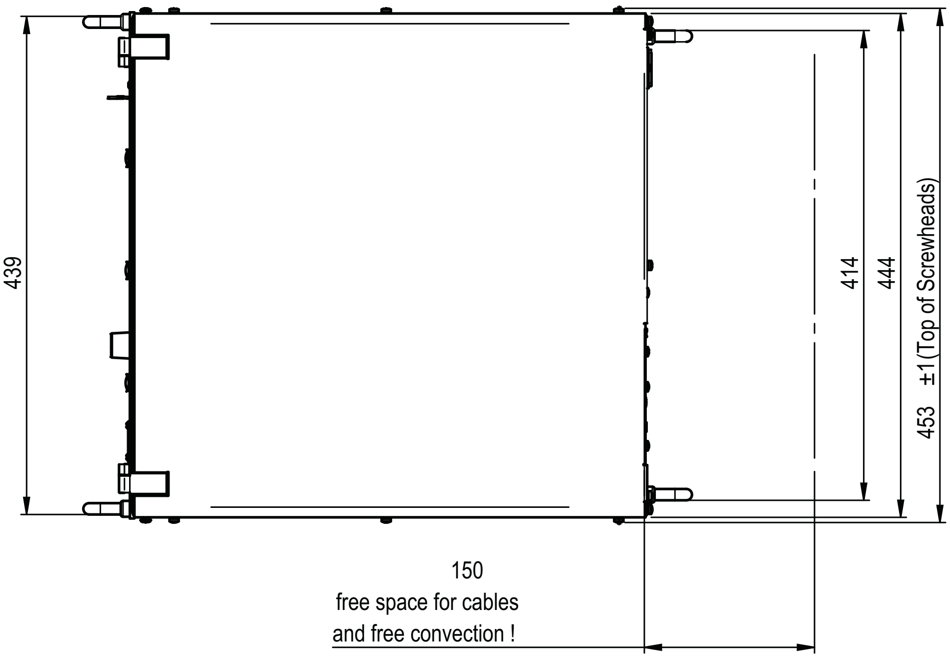


FOOT

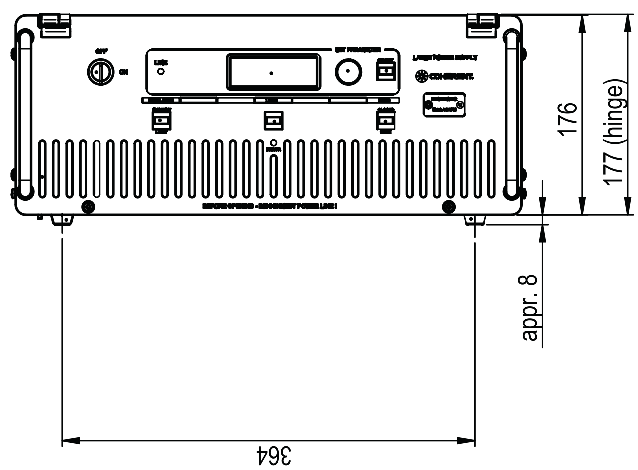
Dimensions are given in (mm).

Figure 2-1. Paladin Scan Head Dimensions (Sheet 3 of 3)

Power Supply



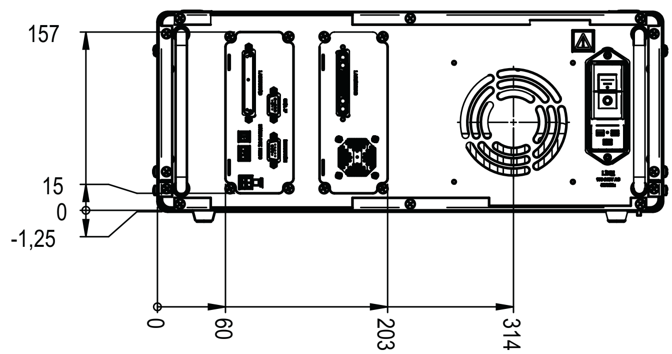
TOP DOWN VIEW



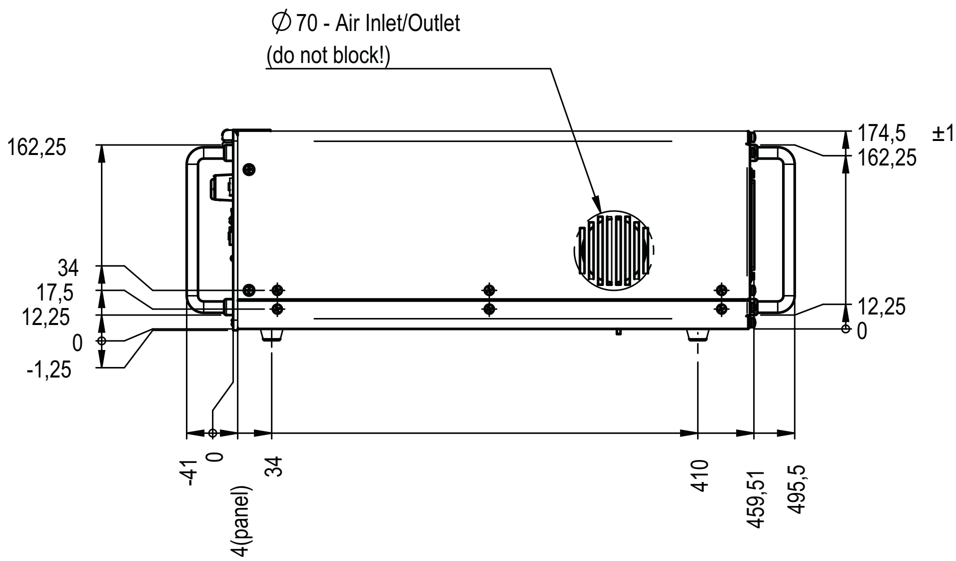
FRONT

Dimensions are given in (mm).

Figure 2-2. Power Supply Dimensions (Sheet 1 of 2)



REAR VIEW



SIDE VIEW

Dimensions are given in (mm).

Figure 2-2. Power Supply Dimensions (Sheet 2 of 2)

SECTION THREE: INSTALLATION

Installation Procedure

To keep the factory warranty, the Paladin Scan Laser must be installed as instructed.

See “Controls, Connectors and Indicators” on page 4-1, for the location of items referred to in the installation procedures. The user must review the information in “Section One: Laser Safety” and research local safety ordinances to verify compliance of regulations prior to installation of the laser system.

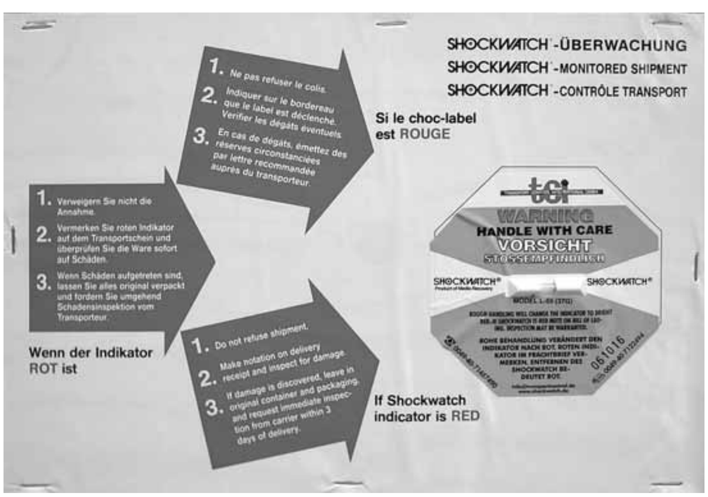
If the power supply is installed on a rack:

1. install the mounting hardware
2. then install the assemblies to the rack

Start the procedures under “System Setup” on page 3-3 only after the assemblies are installed. See “Installation of Rack-Mount Options” on page 3-6 for details.

Shipping Container Inspection

Before beginning the unpacking procedure inspect the “Shock” and “Tilt” watches on the crate (Figure 3-1). If either of these watches have turned red, the shipping company and the local Coherent Service representative must be notified of potential issues.



SHOCK WATCH

TILT WATCH

Figure 3-1. Shipping Label Indicators

Installation Checklist

The following is the Installation Checklist for the Paladin Scan laser system. This checklist is given as a quick reference guide for the experienced end user. Refer to the complete procedure that follows for details on each step.

[ ] Confirm that shock watches have not been activated.

[ ] Remove the laser head and power supply from shipping crate.

[ ] Confirm that all components and accessories are present. Refer to the parts list at the end of this manual.

[ ] Set the laser head on stable surface.

[ ] Set up the power supply and chiller[1] on stable surface.

[ ] Connect and secure the head cables between the power supply and the laser head.

[ ] Connect the purge gas[2].

[ ] Connect the water hoses between the chiller and the laser head. Fill the chiller with recommended coolant[3].

[ ] Connect the power supply to the facility power.

[ ] Connect the chiller to the facility power.

[ ] At the first installation, the laser head and power supply must be thermally acclimatized at ambient temperature to avoid condensation of humidity.

[ ] Turn the main switch on at the back of the power supply.

[ ] Turn the power supply keyswitch to the ON position.

System Unpacking

The system is delivered in two crates:

- power supply
- one for the laser head

The following outlines the procedure for unpacking the system.

---

1. Water chiller is not provided by Coherent and must be provided by the Customer. See "Cooling Water Requirements" on page 2-5.
2. See "Purge Gas Requirements" on page 2-5.
3. AQUA 90 (www.lauda.de/en/)

NOTICE
Keep the shipping container. The container will be required if the system is returned to the factory for service. The container may also be needed to support a shipping damage claim.

Laser Head Shipping Container

1. Remove binding straps from head shipping container.

CAUTION!
The Paladin Scan laser head weights approximately 50 kg (110 lbs). Coherent recommends that multiple personnel lift the laser head from the packaging.

2. Remove the top container cover and packaging material.
3. Lift the packaged laser head from the box. Remove PE packaging.

The procedure described above can be used to unpack the Power Supply.

System Setup

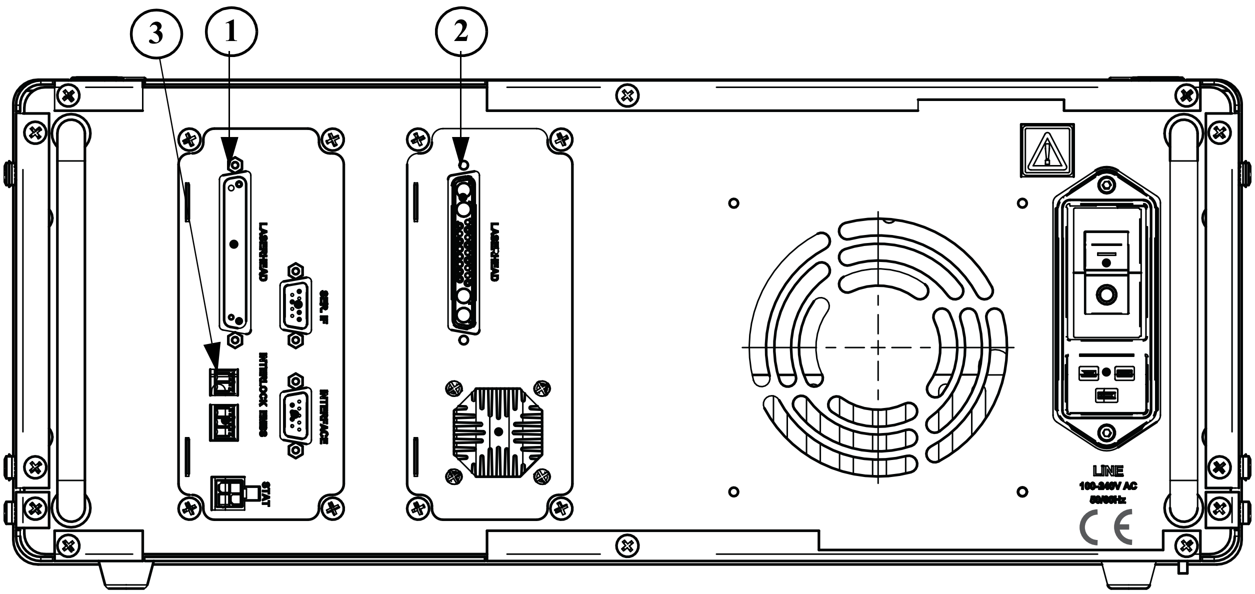
Electrical Connections

1. Plug the 37 pin Sub D master head cable into the laser head and then into the corresponding Sub D connector on the power supply (see Figure 3-2 and Figure 3-3).

1. Head Cable 1 (37-Pin) 2. Head Cable 2

Figure 3-2. System Cables

2. Tighten the small screws on each side of the connector to secure the connectors to the power supply.



1. Head Cable 1 Port

2. Head Cable 2 Port

3. External Interlock

Figure 3-3. System Connections

NOTICE
Accidental disconnection at the laser head or the power supply while the laser is in operation can cause major damage to the system.

3. Connect the laser to an external interlock circuit or install a jumpered interlock connector. See Figure 3-2 for the location of the power supply external interlock connector.

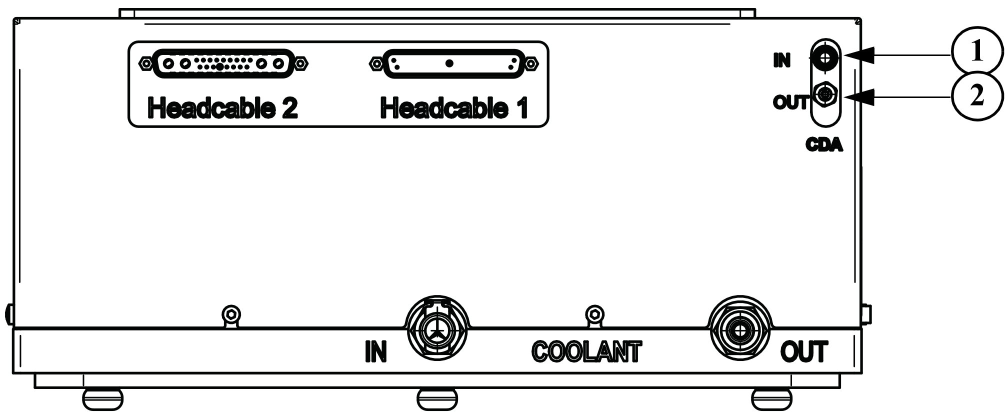
4. When all electrical, purge and water connections are made, use the supplied AC line power cord to connect the power supply to the line voltage.

NOTICE
The head cable must remain connected to the laser head during operation and during the ramp down of the laser.

DANGER!
Wear laser safety eyewear and skin protection to protect against the radiation generated from the laser. It is assumed that the operator has read "Section One: Laser Safety" and understands laser safety practices and the possible danger. Make sure all personnel in the area wear laser safety eyewear.

Clean Dry Air (CDA) Installation

Attach two user provided CDA tubes to the rear of the head and connect to house CDA supply. See Figure 3-4 for location of CDA tube connections. .



1. CDA IN 2.CDA OUT

Figure 3-4. CDA Quick Disconnect

Installation of Rack-Mount Options

The following is the installation procedure for the power supply rack-mount option.

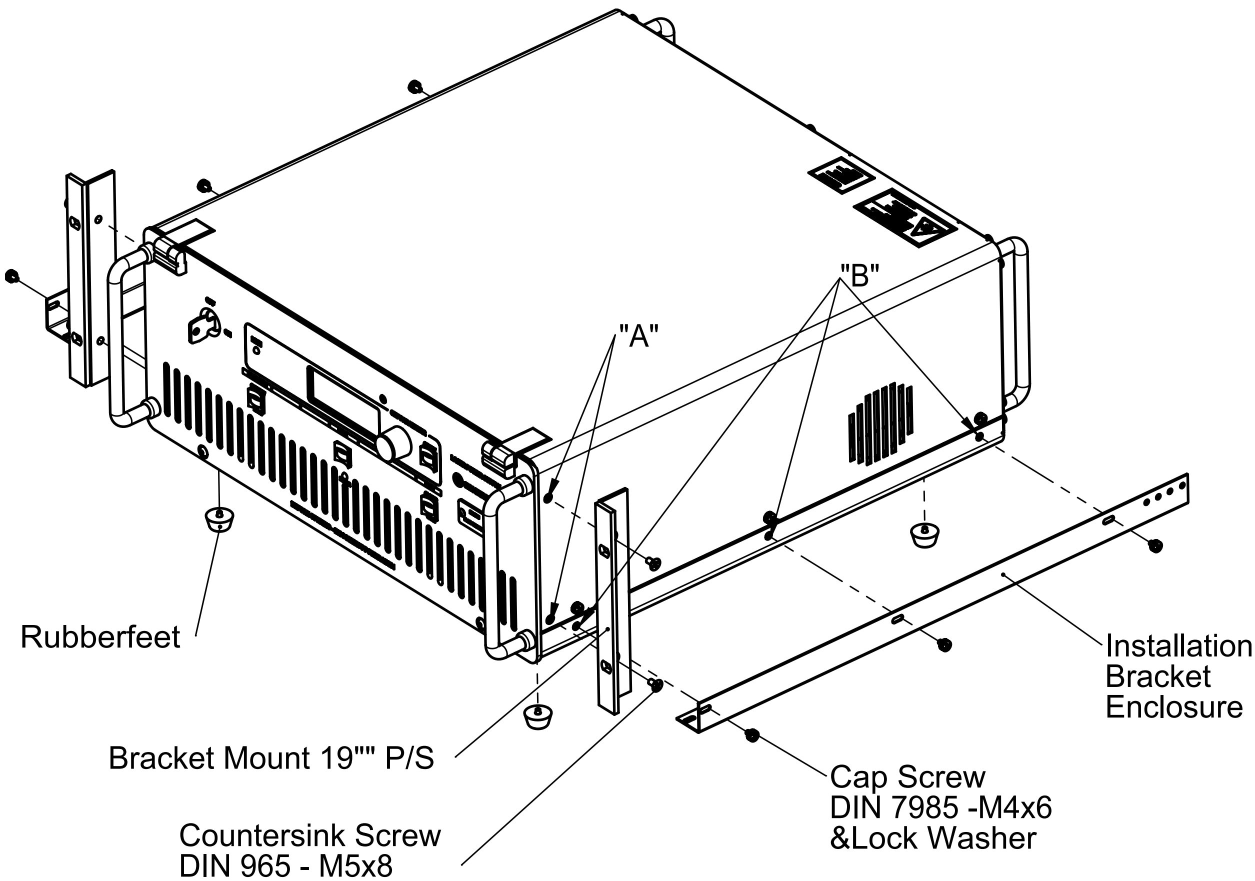


Figure 3-5. Power Supply Rack Mount Option

1. Loosen the 4-cap screws and remove the lock washers from position A on both sides of the power supply (see Figure 3-5).
2. Place the 2 "bracket mount 19" p/s" into position A, on both sides of the power supply. Secure the bracket mount to the power supply with the 4-countersunk screws DIN 965 - M5x8.
3. If the weight of the power supply is to be held by the rack panel, remove the four rubber-feet from the bottom of the power supply.
4. Loosen the 6-cap screws and remove the lock washers from position B.

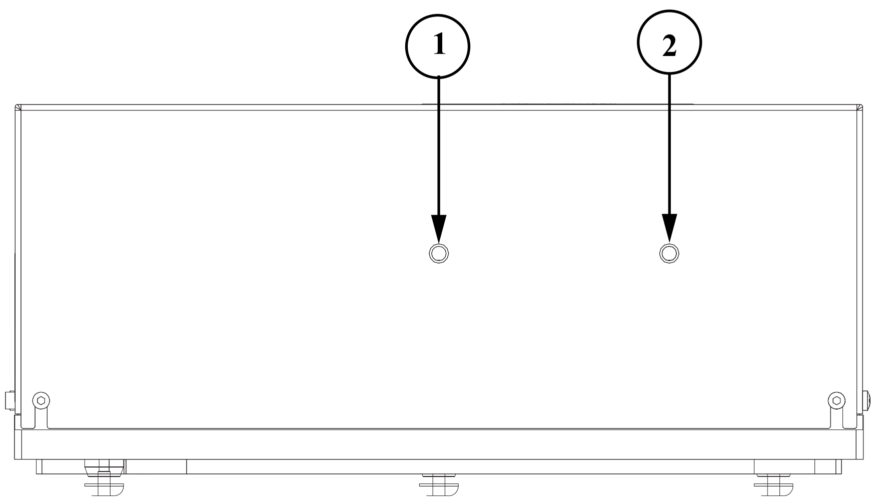
Put the two "installation bracket enclosure" into position B. Secure the enclosure to the power supply with the screws from step 4.

SECTION FOUR: CONTROLS, INDICATORS AND FEATURES

Controls, Connectors and Indicators

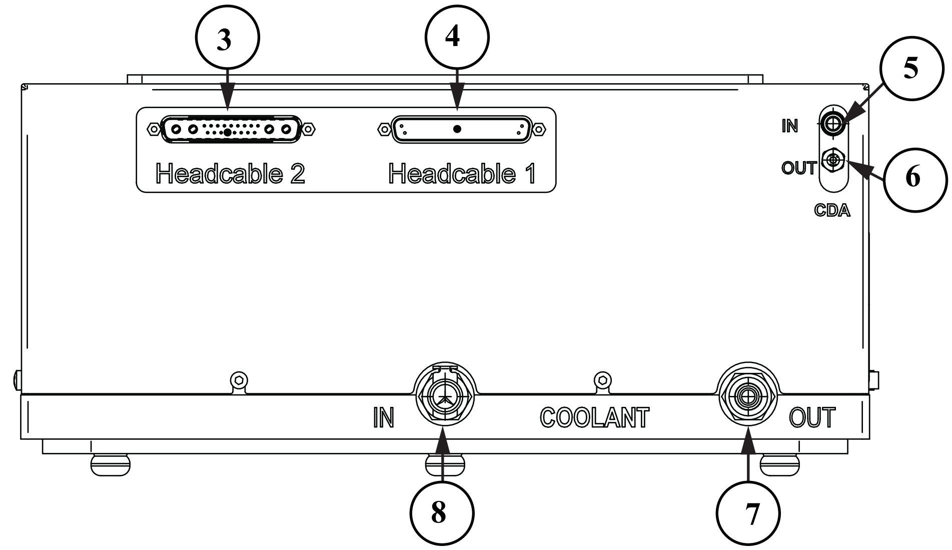
Figure 4-1 and Figure 4-2 provide an overview of the Paladin Scan Modelocked laser system controls, connectors and indicators.

Laser Head



FRONT VIEW

Figure 4-1. Laser Head



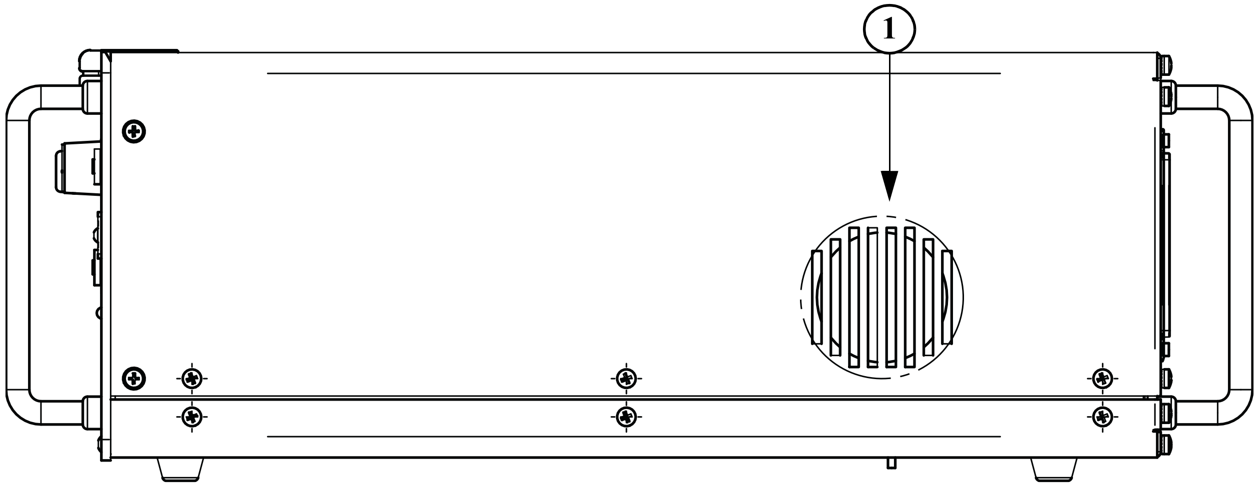
REAR VIEW

TOP VIEW

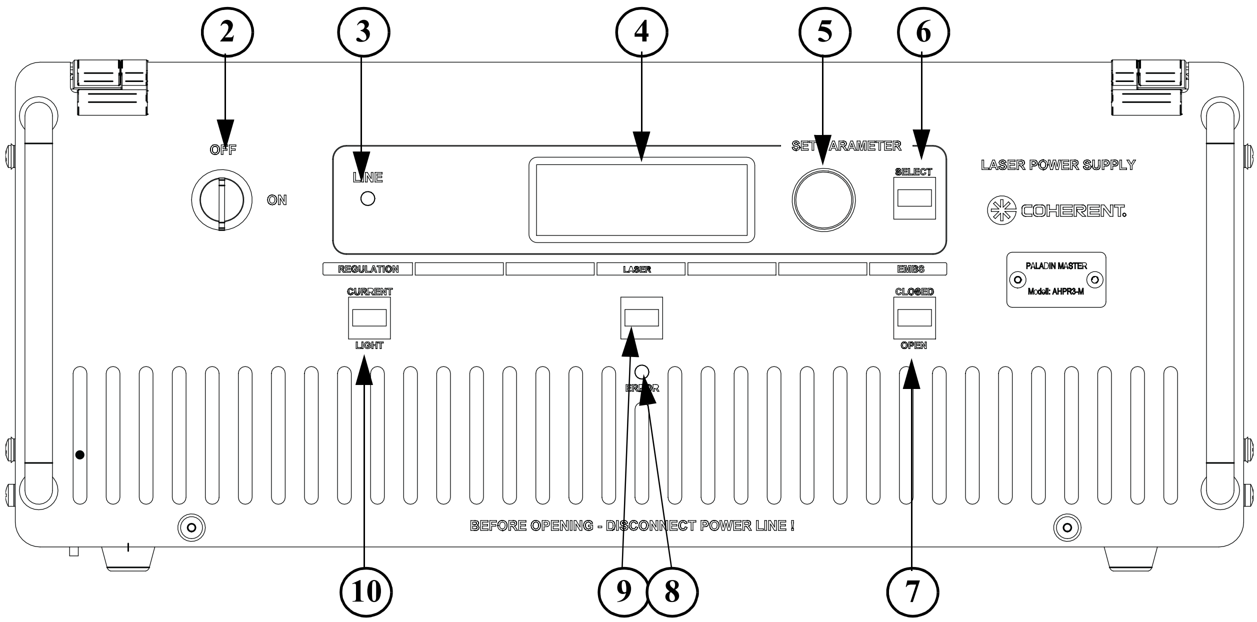
1. 532 nm output window
2. Not in use
3. Head Cable 2 Socket
4. Head Cable 1 Socket (37-pin D-connector)
5. CDA In
6. CDA Out
7. Coolant Out
8. Coolant In
9. Laser Emission Indicator

Figure 4-1. Laser Head (Continued)

Power Supply

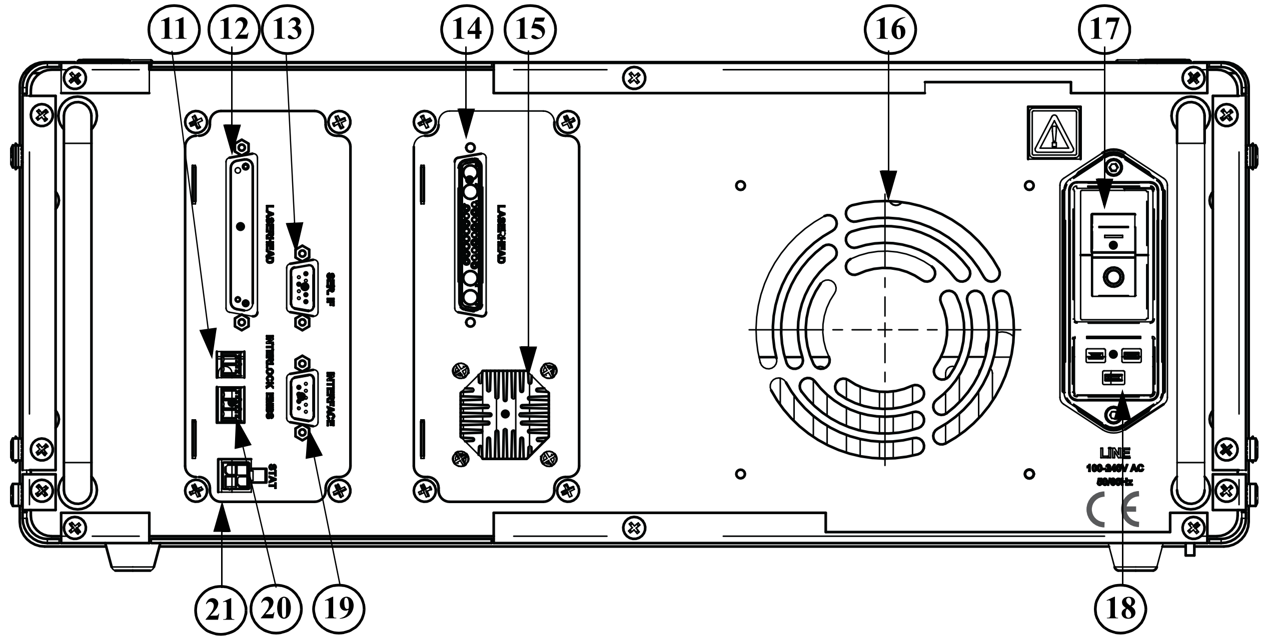


SIDE VIEW



FRONT VIEW

Figure 4-2. Power Supply



REAR VIEW

1. Ventilation
2. Keyswitch
3. Line LED
4. Display
5. Adjustment Knob
6. Select Button
7. Shutter Open/ Close Button
8. Error LED
9. Laser On/Off Button
10. Current/ Light Loop Button
11. External Interlock
12. Head Cable 1 Port
13. RS-232 Serial Interface
14. Head Cable 2 Port
15. Air Outlet
16. Air Outlet
17. Main power switch
18. Mains Inlet
19. Interface (Laser Status Signal)
20. Shutter Interlock (EMBS)
21. Stat (LD-On indicator signal)

Figure 4-2. Power Supply (Continued)

SECTION FIVE: DAILY OPERATION

Introduction

The Paladin Scan Modelocked laser is operated by one power supply. Use the power supply front panel or the computer (PC) using RS-232 interface to control the laser.

NOTICE
Do not make direct changes to the power supply parameters, unless directed by a Coherent support representative.

The laser goes through two sequences before it can function. The sequences are “Warm-up” and “Ramp-Up”. The “Warm-up” sequence raises the temperatures of the crystals to a temperature where the laser can be operated. The “Ramp-Up” sequence starts laser operation.

The laser goes through the “Warm-up” and “Ramp-Up” sequences when the laser is installed and activated for the first time. The laser goes to standby mode when the “Warm-up” sequence is complete. When the laser is in standby mode, only the “Ramp-Up” sequence is necessary.

System Turn-on

Turn-on (Cold Start)

1. Turn ON the Main Power Switch at the power supply rear panel.
2. Turn the key switch to the ON position
3. Wait until the laser is in Standby. See “Initial Warm-Up” on page 5-2 for details.

Initial Warm-Up

Turn the key to the ON position to activate the power supply. The display will show the Warm-up sequence as it goes through 10 individual steps. The 10 steps are shown in Figure 5-1. At screen 9, a

1	*--------Coherent- Revision 3.0060
2	*--------Coherent- ANALYZING HEAD
3	*--------Coherent- ANALYZING HEAD 2
4	*--------Coherent- ANALYZING HEAD 3
5	*--------Coherent- ANALYZING HEAD SBR SHIFER
6	*--------Coherent- ANALYZING HEAD SBR
7	*--------Coherent- ANALYZING LD MODULES
8	*--------Coherent- CHECK SHIFTER
9	Scan-Dual Time left: 4min THERM. EQUILIBRATION
10	POWER (GR) SET 36.0W OSZ 0.0% ACT 0.0W STAND BY

Figure 5-1. Warm-Up Sequence

countdown starts. In the countdown, the display shows the minutes left until thermal equilibration is reached.

Depending on ambient temperature conditions it can take 10 to 45 minutes until the laser reaches Standby status.

After the laser has reached Standby status, laser operation can be activated by pushing the Laser ON/Off button. See "Daily Turn-on (Warm Start)" on page 5-3 for details.

Daily Turn-on (Warm Start)

1. Press the Laser ON button on the front panel of the power supply. The display shows “RAMP UP...” while the laser sets to the requested power. When the laser is ready for operation it will show “NORMAL OPERATION” on the front panel display.

1
POWER (GR) SET 36.0W
OSZ 0.0% ACT 0.0W
STAND BY

2
POWER (GR) MAX 0.0W
OSZ 0.0% ACT 0.0W
RAMP UP OSCILLATOR

3
POWER (GR) MAX 0.0W
OSZ 0.0% ACT 0.0W
RAMP UP OSCILLATOR
RO DETERMINATION

4
POWER (GR) MAX 0.0W
OSZ 85.9% ACT 32.4W
RAMP UP OSCILLATOR
VANADATE TEMP WAIT..

5
POWER (GR) MAX 0.0W
OSZ 85.8% ACT 38.7W
RAMP UP OSCILLATOR
WAIT FOR SHG TEMP

6
POWER (GR) MAX 0.0W
OSZ 85.9% ACT 38.0W
RAMP UP OSCILLATOR
SHG TEMP READY

7
POWER (GR) MAX 0.0W
OSZ 85.1% ACT 36.0W
RAMP UP OSCILLATOR
LIGHTLOOP ENABLED

8
POWER (GR) MAX 0.0W
OSZ 85.1% ACT 36.0W
RAMP UP OSCILLATOR
IR LIGHTLOOP ENABLED

9
POWER (GR) MAX 0.0W
OSZ 85.4% ACT 36.0W
RAMP UP OSCILLATOR
WAIT FOR IR LIGHTLOOP

10
POWER (GR) MAX 0.0W
OSZ 85.4% ACT 36.0W
RAMP UP OSCILLATOR
IR LIGHTLOOP READY

11
POWER (GR) MAX 0.0W
OSZ 85.7% ACT 36.0W
RAMP UP OSCILLATOR

12
POWER (GR) MAX 0.0W
OSZ 85.4% ACT 36.0W
RAMP UP LIGHTLOOP

13
POWER (GR) MAX 36.0W
OSZ 85.7% ACT 36.0W
NORMAL OPERATION

Figure 5-2. Ramp-Up Sequence

2. Press the CLOSED/OPEN button on the power supply front panel to open the shutter.

System Turn-off

Turn-off (Daily Use)

1. Press the CLOSED/OPEN button on the power supply front panel to close the shutter.
2. Press the Laser ON button of the power supply. The display shows "RAMPING DOWN" until the laser goes to standby mode.

Turn-off (Complete Shut-down)

1. Press the CLOSED/OPEN button on the power supply front panel to close the shutter.
2. Press the Laser ON button of the power supply to ramp down the laser.
3. Put the keyswitch of the power supply in the OFF position.

NOTICE
To prevent damage to the Paladin laser system, do not disconnect the power supply from the laser head while laser is on.

System Control

The laser can be controlled with the front panel of the power supply or with RS-232 commands. Figure 4-2 shows the front panel display of the power supply. The "Light Loop/Current mode", "Laser On/Off" and the "Shutter Open/Close" buttons have two yellow LEDs. The LEDs serve as indicators for the actual status of each button.

The power supply display shows the status of the laser. Use the Menu Knob to switch between different displays. The displays show information about the laser system (actual diode currents in percent, operation hours…). Service mode allows the use of the "Light Loop/Current mode" and the "Select" button.

The backside of the power supply has a connector for an external interlock. The external interlock must be connected in order to start the laser system.

Shutter Control

The Paladin Scan is equipped with an Elector-Magnetical Beam Shutter (EMBS). Open and close the shutter with the button on the front panel of the power supply, or with a RS-232 command. The EMBS interlock must be connected to open the shutter. The shutter can only be opened when the laser is in “normal operation” status.

Operating Modes

The Paladin Scan laser has three different modes of operation: “user mode”, “service mode” and “Coherent service mode”. Normal operation will be in “service mode”.

“User Mode”

When the power supply is activated, the laser will automatically go to “user mode” operation. This mode provides laser output at the nominal power level. The power level is stabilized by light loop control. This mode permits only the following actions by the user:

- Laser on/off
- Open or close the shutter
- Check set and actual temperatures (via RS-232)
- Shift SBR to the next available spot (via RS-232)

Service Mode (default)

The service mode allows access to certain commands that are not normally used. These commands affect the operating parameters of the laser and must be used carefully.

Coherent Service Mode

The Coherent service mode is available for Coherent service personnel only. The service mode enables control of all laser parameters. Changing these parameters can damage the laser!

SECTION SIX: COMPUTER CONTROL

General Description

To operate the Paladin Scan Modelocked laser with a computer, use a RS-232 extension cable with a minimum of 3 wires. The pins 2, 3 and 5 must be connected.

Set the RS-232 interface to:

baud-rate	9600
data-bits	8
parity	N
stop-bits	1

Communication Principles

- End each command with a carriage return or a semicolon.
- The firmware starts in Prompt off mode.

Allow the laser approximately 30 minutes to reach thermal equilibrium. During this period some commands are not available. Use the **"?W"** command to check the warm-up status.

In Default Mode (n=0)

The default mode (Command: >= *n* = 0) is recommended for software integration. Each command will be answered by a number or string followed by a colon, a fault number and a semicolon. Example: (command regular text, answer in *italic*):

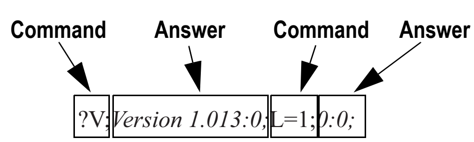


Figure 6-1. Command and Answer Structure in Default Mode (n=0)

In Terminal Mode (n=1)

In terminal mode (n=1) the command will be answered by a number or string in a new line and a fault description again in a new line. Then a prompt will be shown. Example: (command regular text, answer in *italic*)

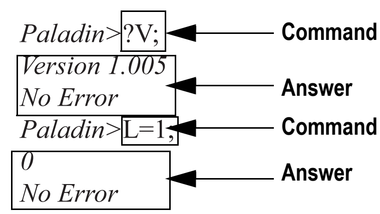


Figure 6-2. Command and Answer Structure in Terminal Mode (n=1)

RS-232 Command Set

User Commands

Table 6-1 and Table 6-2 shows the Paladin user command set. In the tables that follow the letters *n, m* represent digits. Note: Faults are secured in the laser head EPROM and erased when the laser is activated again.Table 6-1 shows basic commands and Table 6-2 shows service commands. The Paladin Scan is set to service mode by default.

Table 6-1. Basic Commands (Sheet 1 of 4)

COMMAND	DESCRIPTION	VALUES
BAUD=*nnnnn*	Sets the baud rate Must have a minimum time of 200ms between sending two commands! The baud rate will go back to default after power off or software reset using the RESET command.	Allowed Baud rates are: 9600 (default) 14400 19200 28800 38400
?W	Returns warm-up status	0 = warm-up 1 = laser is ready to start (nearly 15 min after cold start)
FP=*n*	Disable/ Enable front panel	n=0 disable front panel n=1 enable front panel (default)

Table 6-1. Basic Commands (Sheet 2 of 4)

Command	Description	Values
?FP	Returns front panel status	0 = front panel disabled 1 = front panel enabled
L=*n*	Starting the laser is delayed by a security interval. After that the laser diodes will be switched on and the laser will be ramped up carefully. Check the status to get information when the laser is ready for operation. The laser can be switch on only when the supply is in standby mode.	n=0 switch laser off (default) n=1 switch laser diodes on
?L	Returns actual laser diodes status	0 = laser diodes are off 1 = laser diodes are on
P=*n.m*	Set laser power in Watt Setting the power is only possible in warm-up or standby mode. The power range depends on the laser type.	
?P	Returns actual power in Watt	
?SP	Returns set power in Watt	
S=*n*	The shutter function is available in normal operation mode and alignment mode. The shutter status will be returned.	n=0 close beam shutter (default) n=1 open beam shutter
?S	Returns actual shutter status	0 = shutter is closed 1 = shutter is open
D=*n*	Turns display LEDs on or off.	n=0 set display LED’s off n=1 set display LED’s on
?OPH*n*	Returns actual operating hours	n=0 laser head n=1 diode oscillator n=3 diode amplifier n=9 power supply
?SN*n*	Returns the serial numbers	n=0 laser head n=1 diode oscillator n=3 diode amplifier n=9 power supply
?SPHSBR	Returns operating hours of the actual spot on SBR.	
?DT*n*	Returns actual temperatures of the diodes	n=1 diode oscillator n=3 diode amplifier

Table 6-1. Basic Commands (Sheet 3 of 4)

COMMAND	DESCRIPTION	VALUES
?T*n*	Returns actual crystal temperatures	n=0 vanadate oscillator n=1 resonator n=2 vanadate amplifier n=5 doubler n=7 SBR
?ST*n*	Returns crystal set temperatures	n=0 vanadate oscillator n=1 resonator n=2 vanadate amplifier n=5 doubler n=7 SBR
?HT*n*	Returns heatsink temperatures	n=0 heatsink laser head n=1 heatsink PS
?V	Returns software version	
?LT	Returns laser type	
?CIP	Returns actual oscillator current in percent	
?CIPA	Returns actual amplifier current in percent	
?SCIP	Returns oscillator set current in percent	
?SCIPA	Returns amplifier set current in percent	
?STA	Returns operating status	0: Ramp-up oscillator 1: Equilibration oscillator 2: Checking IR Oscillator 3: --- 4: --- 5: --- 6: --- 7: Ramp-up Lightloop 8: Normal operation 9: --- 10: Ramp down oscillator 11: Standby 12: Warm-up 13: Alignment mode wait 14: Alignment mode ready
?LF	Returns the last 10 fault numbers and related laser head hours	

Table 6-1. Basic Commands (Sheet 4 of 4)

COMMAND	DESCRIPTION	VALUES
CF	Clears the fault buffer	
?PRO	Returns the roll over power at last start up in MAINT=1	
?IRO	Returns the current in percent at roll over power at last start up in MAINT=1	

Service Commends

Table 6-2. Service Commands (Sheet 1 of 3)

COMMANDS	DEFINITION	VALUES
SM=*n*	This command can be used only at laser status 11 = standby or 12 = warmup (see command ?STA)	n=0 sets supply to User Mode n=1 sets supply to Service Mode
?SM	Returns service status	0 = User Mode 1 = Service Mode
The following commands are available only in service mode!		
~	Sending the ~ character will reset the RS-232 communication. This command needs no limiter like all other commands. The input and output buffer will be reset to start conditions. ~ characters within a command will be ignored.	Examples: Send: ?V; Receive: 3.0000:0; Send: ?V;~ Receive: Send:~?V; Receive: 3.0000:0; Send:?~V; Receive: 3.0000:0;
NP=*n*		n=0 communication protocol Paladin compatible (default) Send: ?V; Receive: 3.0000:0; n=1 communication protocol with additional command string. Send: ?V; Receive: ?V:3.0000:0;

Table 6-2. Service Commands (Sheet 2 of 3)

COMMANDS	DEFINITION	VALUES
AM=*n*	The AM command can be used only when the laser is on and laser status is: 8 = Normal operation or 14 = Alignment Mode ready When switching to alignment mode the laser will need 4 to 5 minutes to get stable (status = 13 alignment mode wait). Then it will go to status 14 = alignment mode ready. The shutter function is available in status 8 and 14. Switching back to normal mode will take 4 to 5 minutes.	n=0 sets laser to normal mode (default) n=1 sets laser to alignment mode
?AM	Returns alignment mode status	
AP+	Adjust internal Green-detector	
AP-	Adjust internal Green-detector	
AP0	Reset adjustment to factory settings	
?AP	Query Green-detector adjustment	
APIR+	Adjust internal IR-detector	
APIR-	Adjust internal IR-detector	
APIR0	Reset adjustment to factory settings	
?APIR	Query IR-detector adjustment	
SWT=*mnnnnn*	Set warm-up timer to nnnnn seconds.	m=1 Warm-up timer m=2 Ramp-up timer
?SWT*m*	Returns number of seconds left till warm-up is complete.	m=1 Warm-up timer m=2 Ramp-up timer
?SWTSET*m*	Returns set value of warm-up timer m.	m=1 Warm-up timer m=2 Ramp-up timer
MAINT=*n*	Set maintenance level to n. n=0..4 The ramp-up procedures are defined by eprom parameters. The maintenance level should be set in Standby mode and need to be reset after ramp-up manually. Switching the power supply off will set back the level to standard 0.	n=0 Standard ramp-up n=1 Roll over determination n=2 Not used n=3 Not used n=4 Not used
?MAINT	Returns the maintenance level	

Table 6-2. Service Commands (Sheet 3 of 3)

COMMANDS	DEFINITION	VALUES
?EMBS	Returns EMBS loop status	0 = unknown shutter is closed 1 = closed shutter is open 2 = shutter has been closed by loop
TF=	If the transport flag is set, the time to standby status will be about 3 times longer than normal. The 2f temperatures will be ramped-up 3 times slower. The flag will be reset, if the status goes from warm-up to standby. The information is still present in a system variable and can be used during current ramp-up to decide whether a optimization will be done or not. Finally the system variable will be reset in normal operation.	0 = Reset transport flag 1 = Set transport flag
?TF	Returns transport flag status	0 = transport flag not set 1 = transport flag is set

Power Control Commands

Table 6-3. Power Control Command

COMMAND	DEFINITION	VALUES
SLL=	Sets the lower threshold for Green-Power	
?LL	Returns the lower threshold for Green-Power	
SHL=	Sets the upper threshold for Green-Power	
?HL	Returns the upper threshold for Green-Power	
?FL	Returns the fault level for Green-Power	
?WR	Returns the warning flag.	0 = no warning. Green-Power is in chosen range 1 = warning. Green-Power was out of range.
CWR	Clears the warning flag Green and IR	

Current Mode Related Commands

Table 6-4. Current Mode Related Commands

COMMAND	DEFINITION	VALUES
CM=*n*	Sets regulation mode. This command ca be used in standby only	n=0 Light loop n=1 Current mode
?CM	Returns the regulation mode	0= Light loop 1= Current mode
CIP=nnn.m	Set oscillator current in percent in current mode	
CIPA=knnn.m	Set amplifier current in percent in current mode	Only k=0 available

SBR Related Commands

Table 6-5. SBR Related Commands (Sheet 1 of 2)

COMMAND	DEFINITION	VALUES
?SBRLIM	Returns Life time per SBR spot.	
?SBRMAX	Returns maximum number of SBR spots.	
?SBRH*nn*	Returns spot hours of SBR spot nn	
?SBRST*nn*	Returns spot status of SBR spot nn	0 = good 1 = used 2 = bad 4 = in use
?SPNSBR	Returns the number of the current SBR spot	
SHSBR*m*	Shift SBR crystal to new position	m = 0 mark spot as used m = 1 mark spot as bad Used spot cannot be shifted to, if the spot hours are greater than the maximum spot life time.

Table 6-5. SBR Related Commands (Sheet 2 of 2)

COMMAND	DEFINITION	VALUES
SHSBRTO=*mnn*	Shift SBR to spot number nn. Used spot cannot be shifted to, if the spot hours are greater than the maximum spot life time.	m = 0 mark spot as used m = 1 mark spot as bad
?SBRALL	Returns all SBR spot data. All parameters are separated|.The format is: Maxspots Maximum number of SBR spots X Number of spots in direction X Y Number of spots in direction Y Status1 Status of spot number 1 Hours1 Operating hours of spot number 1 Status2 Status of spot number 2 Hours2 Operating hours of spot number 2 .. Statusn Status of spot number n with n = X times Y Hoursn Operating hours of spot number n with n = X times Y The status code is the same as ?SBRSTnn	
?SBRTLIM	Returns SBR temperature limit	
SBRTLIM=	Set SBR temperature limit. Exceeding the limit will cause a warning.	
?SBRWR	Returns SBR temperature warning flag.	0 = no warning. SBR temperature ok. 1 = warning. SBR temperature above limit.
SBRCWR	Clears SBR temperature warning flag	
SBRRSST=*nn*	Reset SBR spot status of spot nn from BAD to USED	

SHG Related Commands

Table 6-6. SHG Related Commands

COMMAND	DEFINITION	VALUES
DBL=*n*	Disable/Enable doubler optimization	n=0 Disable Doubler optimization n=1 Enable Doubler optimization
?DBL	Returns status of doubler optimization	0 = Doubler optimization disabled 1 = Doubler optimization enabled
RESET	Software reset of the power supply	
GOON	Ignore warm-up, goes to standby	

Temperature Settings Commands

Table 6-7. Temperatures Settings Commands

COMMAND	DEFINITION	VALUES
SETVAN=*nm.k*	m.k Temperature in °C maximum 6 digits	n=0 Vanadate Oscillator n=1 Resonator plate n=2 Vanadate Amplifier
SETSHG=*m.k*	m.k SHG-Temperature in °C maximum 6 digits	

Monitoring Commands

Table 6-8. Monitoring Commands (Sheet 1 of 2)

COMMAND	DEFINITION	VALUES
?KTALLS	Returns all relevant data for monitoring. All parameters are separated.	
		Serial number laser head
		Serial number power supply
		Serial number oscillator laser diode
		Serial number amplifier laser diode
		Software version
		Oscillator diode temperature
		Amplifier diode temperature
		Oscillator vanadate temperature
		Amplifier vanadate temperature
		Resonator temperature
		SHG temperature
		SBR temperature
		Heat sink head temperature
		Heat sink power supply temperature
		Oscillator diode current
		Amplifier diode current
		Oscillator diode current (percent)
		Amplifier diode current (percent)
		Laser power green
		Operating hours laser head
		Operating hours Oscillator laser diode
		Operating hours Amplifier laser diode

Table 6-8. Monitoring Commands (Sheet 2 of 2)

COMMAND	DEFINITION	VALUES
?KTALLS	Returns all relevant data for monitoring. All parameters are separated.	
		Operating hours power supply
		Laser operating status
		Regulation status
		Laser-ready bit status
		Roll over power green
		Roll over current (percent)
		Warm-up timer
		Ramp-up timer
		SBR spot number
		Spot hours SBR
		Warning flag
		SBR Warning flag
		Laser power fault level green
		Hardware fault: ?HWF
		Most recent fault
		Most recent fault related head hours
		Opt. status
		Current fault text

Settings

Table 6-9. Settings Commands

COMMAND	DEFINITION	VALUES
?SETTINGS	Returns all set values for the specific model. All parameters are separated.	
		Laser type
		Oscillator diode set temperature
		Amplifier diode set temperature
		Oscillator vanadate set temperature
		Amplifier vanadate set temperature
		Resonator set temperature
		SHG set temperature
		Oscillator laser diode set current (percent)
		Amplifier laser diode set current (percent)
		Laser set power green
		Maximum current oscillator diode (percent)
		Maximum current amplifier diode (percent)
		Laser power low level green
		Laser power high level green

SECTION SEVEN: MAINTENANCE AND TROUBLESHOOTING

Faults and System Messages

All faults or system messages given by the Paladin Scan software are shown on the system display and can be handled by RS-232 commands. These faults and messages are listed in Table 7-1.

When a fault condition occurs the diodes and emission indicators will turn off. The power supply display and the RS-232 interface will indicate a fault condition has occurred. To return the system to normal operation, clear the fault.

Table 7-1. Fault List (Sheet 1 of 3)

#	Message in Display	Message via RS-232	Troubleshooting	Reason
1	Err NTC Heats short	NTC Heat sink Laser-diodes Master shorted!	Coherent Service	-
2	Err NTC Heats short	NTC Heat sink Laser-diodes Master broken!	Coherent Service	-
3	Err TEMP Heats Hi	Temperature Heat sink Laser-diodes too high!	Coherent Service	T > 55°C
13	Err NTC Van1 shorted	NTC Vanadate 1 shorted!	Coherent Service	-
14	Err NTC Van1 broken	NTC Vanadate 1 broken!	Coherent Service	-
15	Err TEMP Van1 Hi	Temperature Vanadate 1 too high!	Coherent Service	T > 45°C
16	Err NTC Res shorted	NTC Resonator shorted!	Coherent Service	-
17	Err NTC Res broken	NTC Resonator broken!	Coherent Service	-
18	Err TEMP Res Hi	Temperature Resonator too high!	Coherent Service	T > 45°C
19	Err NTC SBR shorted	NTC SBR shorted!	Coherent Service	-
20	Err NTC SBR broken	NTC SBR broken!	Coherent Service	-
21	Err TEMP SBR Hi	Temperature SBR too high!	Coherent Service	T > 70°C
25	Err NTC Dbl shorted	NTC Doubler shorted!	Coherent Service	-
26	Err NTC Dbl broken	NTC Doubler broken!	Coherent Service	-

Table 7-1. Fault List (Sheet 2 of 3)

#	Message in Display	Message via RS-232	Troubleshooting	Reason
27	Err TEMP Doubler Hi	Temperature Doubler too high!	Coherent Service	T > 170 °C
28	Err NTC HtsH shorted	NTC Heat sink Laser head shorted!	Coherent Service	-
29	Err NTC HtsH broken	NTC Heat sink Laser head broken!	Coherent Service	-
30	Err TEMP HtsH Hi	Temperature Heat sink laser head too high!	Coherent Service	T > 55 °C
31	Err Interlock active	Interlock loop open!	Coherent Service	-
32	Err Beamshutter	Beam-shutter fault detected!	Coherent Service	-
33	Err Over Current LD1	Over Current Laser-diode 1!	Coherent Service	Depends on diode
41	Err NTC LD1 shorted	NTC Laser-diode 1 shorted!	Coherent Service	-
42	Err NTC LD1 broken	NTC Laser-diode 1 broken!	Coherent Service	-
44	Err TEMP LD1 Hi	Temperature Laser-diode 1 too high!	Coherent Service	T > 35 °C
79	Err EPR Head defect	Head EPROM defect!	Coherent Service	-
80	Err EPR Cryst defect	Shifter EPROM defect!	Coherent Service	-
81	Err EPR Supply defect	Supply EPROM defect!	Coherent Service	-
82	Err EPR LD1 defect	Laser-diode 1 EPROM defect!	Coherent Service	-
90		Checksum incorrect!	Coherent Service	-
91		Checksum is zero!	Coherent Service	-
92	Err Ld's Osc down	Laser-diodes Oscillator down!	Coherent Service	-
105	Err VAN1 Temp wrong	Vanadate Temperature out of range!	Coherent Service	$\Delta T > 1$ K
106	Err RES Temp wrong	Resonator Temperature out of range!	Coherent Service	$\Delta T > 1$ K
110	Err 2F Temp wrong	Doubler Temperature out of range!	Coherent Service	$\Delta T > 1$ K
113	Err Security LED's	Head security LED's defect! Shutter intentionally closed!	Coherent Service	-

Table 7-1. Fault List (Sheet 3 of 3)

#	Message in Display	Message via RS-232	Troubleshooting	Reason
114	Err I2C-Bus 1	Communication error I2C-Bus 1!	Coherent Service	-
115	Err I2C-Bus 2	Communication error I2C-Bus 2!	Coherent Service	-
116	Err I2C-Bus 3	Communication error I2C-Bus 3!	Coherent Service	-
117	Err I2C-Bus 4	Communication error I2C-Bus 4!	Coherent Service	-
121		Oscillator current above limit. System halted!	Coherent Service	I > 110 %
122		Amplifier current above limit. System halted!	Coherent Service	I > 110 %
124		Head EPROM 2 not available!	Coherent Service	-

Table 7-2. System Messages (Sheet 1 of 3)

#	MESSAGE	TROUBLESHOOTING
0	OK	--
77	Bad data format!	Section 3
78	Range error!	Section 3
96	Shifting not allowed!	Coherent Service
118	Adjust is at lower limit!	Section 4
119	Adjust is at upper limit!	Section 4
120	Allowed only in Standby!	Section 5
123	UV power is out of range!	Coherent Service
125	Time to shift to next spot!	--
126	Last spot in use!	--
130	Allowed only in normal operation!	--
131	UV-power not reliable in alignment mode!	--
132	Alignment mode not available	--
134	Output power ok! Power headroom < 5%	--

Table 7-2. System Messages (Sheet 2 of 3)

#	MESSAGE	TROUBLESHOOTING
135	Chiller fault!	--
136	Head humidity too high!	--
140	Internal fault Slave!	--
143	Internal fault master!	--
150	Laser head cover is open!	--
151	Allowed only in Standby/Warmup!	--
152	SBR fault!	--
156	LD's unintensionally enabled!	--
157	Watchdog doesn't work!	--
159	Beamshutter always open!	--
160	Beamshutter communication fault!	--
161	Beamshutter defect! Wrong status!	--
162	Flow control fault in security loop!	--
163	Plausibility fault in security loop!	--
164	Flash Memory fault!	--
165	Laser head not supported!	--
166	Laser - Laser diode mismatch!	--
167	Laser diode wavelength mismatch!	--
168	Battery is low!	--
169	Change Baudrate not supported!	--
170	Wrong Baudrate!	--
171	Common Service Interface not enabled!	--
172	Amplifier diodes inconsistent!	--
173	Laser warm-up too long!	--
174	Laser ramp-up too long!	--
175	Mismatch number of PS's and LD's!	--
176	SBR temperature above limit!	--
177	Time to shift to next SBR spot!	--

Table 7-2. System Messages (Sheet 3 of 3)

#	MESSAGE	TROUBLESHOOTING
178	Shifter in use	--
179	Last SBR spot in use	--
180	Check ignored	--
181	CDA FLOW OFF	--
182	HEAD LIFE TIME	--
183	LD LIFE TIME	--
184	PSLIFE TIME	--
186	SHG OPT DUE	--
190	SBR TIME LIMIT	--
191	Not allowed to set THG/FHG temperature	--
193	Shifter Alignment done	--
194	Shifter out of range	--
201	Shifter fault	--
202	Shifter loop open	--
203	Shifter at limit	--
204	Shifter operation fault	--
205	Shifter power amplifier fault	--
206	Battery is defect	--
208	Shifter operation fault	--
219	Spot Time Limit	--
221	Plausibility Check Incorrect	--

Preventative Maintenance (PM) Schedule

Table 7-3 lists the recommended PM schedule for trained integrator visits. The schedule is based on each individual system component. Maintenance procedures that can be performed by the operator for example:

- "Replacing the Power Supply"

Can be found in the subsections after the table. Full detail on the PM procedures is shown in the Paladin Service manual.

Table 7-3. Maintenance Schedule

COMPONENT	SERVICE TASK	SERVICE CYCLE
LASER HEAD	Output Window Rotation	As Required
	Recalibration of output power	Every 6 Months
	Head Refurbishment	20,000h (Recommendation)
POWER SUPPLY	Software Update	As Required
PUMP MODULE	Pump Module Replacement	20,000h (Recommendation)

Replacing the Power Supply

1. Turn the key switch to the OFF position.
2. Disconnect the facility power cord after.
3. Disconnect the communication cables between the power supply and the laser head.

To connect the components again, start with the power supply-to-head cables followed the facility power cords. Refer to "Section Three: Installation" on page 3-1 for a complete description on the installation of the Paladin Scan laser system.

Appendix A: Parts List

The following parts can be ordered by contacting our regional Technical Support Groups. Visit our website for contact information in your region.

https://www.coherent.com/support/

When communicating with our Technical Support Department, via telephone or E-mail, the model and laser head serial number of your laser system will be required by the Support Engineer responding to your request.

Table A-1. Parts List

Part Number	Part Description
1282969	Paladin Scan 532-36W
1278533	Paladin Scan/ Dual Power Supply RoHS
1314788	Paladin SCAN/DUAL Head Cable Set 5 m (contains 1 ea. head cable #1 and #2)
1189652	Quick disconnect coupling set (1 x male, 1 x female) cleaned packaged
1312209	Paladin SCAN/DUAL Pump Module Assembly
1284016	Paladin High Power Shutter 1064/532nm
1164557	Paladin SCAN/DUAL Replacement Desiccant
1314664	Paladin SCAN/DUAL Shipping Box Laser Head
1314665	Paladin SCAN/DUAL Shipping Box Power Supply

Appendix B: Deinstallation and Packing Procedure

Deinstallation Procedure

The following instructions give a step-by-step procedure for deinstalling and packing the Paladin laser system.

System Shutdown

Prior to deinstallation, the laser system will need to be shutdown:

1. Press the LASER ON/OFF button on the power supply. When the display indicates STANDBY mode and the LASER ON/OFF button stops blinking, turn the keyswitch to the OFF position.
2. Disconnect the head cables between the power supply and the laser head.

Deinstallation Checklist

The following is the Deinstallation Checklist for the Paladin laser system. This checklist is given as a quick reference guide for the experienced field engineer. Refer to the complete procedure that follows for details on each step.

[ ] Place the system in Standby.

[ ] Turn the keyswitch to the OFF position.

[ ] Press the “Laser On/Off” button on the power supply to turn the laser off.

[ ] Disconnect CDA

[ ] Disconnect the power supply from the facility power.

[ ] Disconnect the head cable from laser head and the power supply.

[ ] Remove all cables and connectors from power supply.

[ ] Turn off the chiller.

[ ] Remove the chiller hoses.

[ ] Empty coolant from laser head, chiller and hoses.

[ ] Carefully pack the laser head and power supply.

Packing the System

Only use Coherent supplied containers for the packing and shipment of the Paladin laser system. If original containers are missing, contact Coherent Service to get replacements.

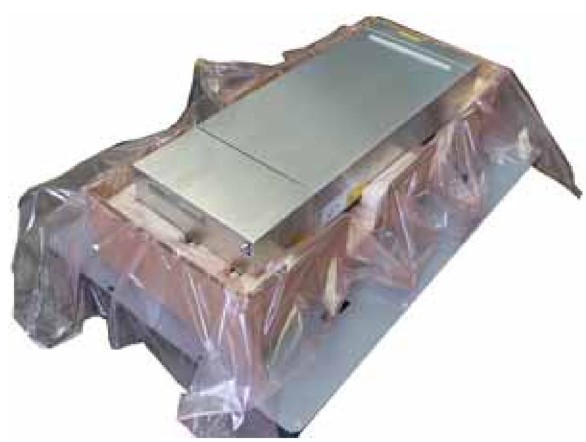
DANGER!
Two person lift required to put the laser head into the crate correctly!

Laser Head Packing

The following procedure describes the packaging requirements for the laser head.

1. Place PE packaging into the laser head box.
2. Place the laser head into the laser head box.
3. Warp and tape the PE packaging around the laser head.
4. Place foam packaging on over the laser head.
5. Place cable accessories in foam packaging
6. Cover the foam packaging.
7. Place box top over the laser head box.
8. Strap the box top to the laser head box.
9. Place Serial Number, Shock Watch and Tilt Watch on the laser head box.

PE Packaging in Laser Head Box



Laser Head in Laser Head Box

Figure B-1. Laser Head Packaging

Taped PE Packaging

Foam Packaging

Cable Accessories in Foam Package

Foam Packaging Cover

Figure B-1. Laser Head Packaging (Continued)

Box Top

Strapped Laser Head Box

Shock Watch

Tilt Watch

Paladin SCAN™
355-35000
LDP.xxxxxxx.xxxxxx

Serial Number Sticker Label

Figure B-1. Laser Head Packaging (Continued)

Power Supply Packing

The following procedure describes the packaging requirements for the power supply.

1. Place PE packaging into the power supply box.
2. Place the laser head into the power supply box.
3. Warp and tape the PE packaging around the power supply.
4. Place foam packaging on over the power supply.
5. Place box top over the laser head box.
6. Strap the box top to the laser head box.
7. Place Serial Number, Shock Watch and Tilt Watch on the laser head box.

PE Packaging in Power Supply Box   Power Supply in Power Supply Box

Figure B-2. Power Supply Packaging

Taped PE Packaging

Foam Packaging

Top Box Cover with Straps

Serial Number Sticker Label

Figure B-2. Power Supply Packaging (Continued)

WARRANTY

Coherent, Inc. warrants the Paladin Scan laser systems to only the original purchaser (the Buyer) and that the laser system, that is the subject of this sale, (a) conforms to Coherent's published specifications and (b) is free from defects in materials and workmanship.

Laser systems are warranted to conform to Coherent's published specifications and to be free from defects in materials and workmanship for a period twenty-four (24) months or 8760 hours f operation, whatever occurs first. Replacement units shipped within warranty will carry the remainder of the original unit’s warranty.

Responsibilities of the Buyer

The buyer is responsible for providing the appropriate utilities and an operating environment as outlined in the product literature. Damage to the laser system caused by failure of buyer's utilities or failure to maintain an appropriate operating environment is solely the responsibility of the buyer and is specifically excluded from any warranty, warranty extension, or service agreement.

The Buyer is responsible for prompt notification to Coherent of any claims made under warranty. In no event will Coherent be responsible for warranty claims made later than seven (7) days after the expiration of the warranty period.

Limitations of Warranty

The foregoing warranty shall not apply to defects resulting from:

- Components and accessories manufactured by companies, other than Coherent, which have separate warranties
- Improper or inadequate maintenance by the buyer
- Buyer-supplied interfacing
- Operation outside the environmental specifications of the product
- Unauthorized modification or misuse
- Improper site preparation and maintenance
- Opening the housing

Coherent assumes no responsibility for customer-supplied material. The obligations of Coherent are limited to repairing or replacing, without charge, equipment that proves to be defective during the warranty period. Replacement sub-assemblies may contain reconditioned parts. Repaired or replaced parts are warranted for the duration of the original warranty period only. The warranty on parts purchased after expiration of system warranty is ninety (90) days. The warranty does not cover damage due to misuse, negligence or accidents, damage due to installations, repairs or adjustments not specifically authorized by Coherent.

The warranty applies only to the original purchaser at the initial installation point in the country of purchase, unless otherwise specified in the sales contract. The warranty is transferable to another location or to another customer only by special agreement that will include additional inspection or installation at the new site. Coherent disclaims any responsibility to provide product warranty, technical or service support to a customer that acquires products from someone other than Coherent or an authorized representative.

THIS WARRANTY IS EXCLUSIVE IN LIEU OF ALL OTHER WARRANTIES, WHETHER WRITTEN, ORAL OR IMPLIED, AND DOES NOT COVER INCIDENTAL OR CONSEQUENTIAL LOSS. COHERENT SPECIFICALLY DISCLAIMS THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

GLOSSARY

°C	Degrees centigrade or Celsius
°F	Degrees Fahrenheit
µ	Micron(s)
µm	Micrometer(s) = 10^{-6} meters
µrad	Microradian(s) = 10^{-6} radians
µsec	Microsecond(s) = 10^{-6} seconds
$1/e^2$	Beam diameter parameter = 0.13534
AC	Alternating current
ADC	Analog-to-digital converter
Amp	Ampere(s)
CDA	Clean Dry Air
CDRH	Center for Devices and Radiological Health
cm	Centimeter(s)
CPU	Central processing unit
DC	Direct current
EEPROM	Electrically erasable programmable read only memory
EMI	Electro Magnetic Interface
g	Gram(s) or earth's gravitational force (gravity)
GND	Ground
HCl	Hydrochloric acid
Hz	Hertz or cycles per second (frequency) (= 1/pulse period)
IR	Infrared (wavelength)
kg	Kilogram(s) = 10^3 grams
KHz	Kilohertz = 10^3 hertz
Kohm	Kilohm(s)
LD	Laser diode
LED	Light emitting diode
m	Meter(s) (length)
mA	Milliamp(s) = 10^{-3} Amperes
mAmp	Milliampere(s)
MHz	Megahertz = 10^6 hertz
mm	Millimeter(s) = 10^{-3} meters
mrad	Milliradian(s) = 10^{-3} radians (angle)
ms	Millisecond(s) = 10^{-3} seconds
mV	Millivolt(s)
mW	Milliwatt(s) = 10^{-3} Watts (power)
NH_3	Ammonia

NH_4	Ammonium
nm	Nanometer(s) = 10^{-9} meters (wavelength)
NMHC	Nitrogen Oxides
OEM	Original equipment manufacturer
PCB	printed circuit board
ps	Picosecond = 10^{-12} seconds
rms	Root mean square (effective value of a sinusoidal wave)
RXD	Receive data
SOX	Sulfur Oxides
TEC	Thermo-electric cooler
TEM	Transverse electromagnetic mode (cross-sectional laser beam mode)
TTL	Transistor-to-transistor logic
TXD	Transmit data
V	Volt(s)
VAC	Volts, alternating current
VDC	Volts, direct current
VECSEL	Vertical External Cavity Surface Emitting Laser
W	Watt(s) (power)

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Paladin Scan Modelocked Laser System Operator's Manual

Part No. 1298483 Rev. AA

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Superior Reliability & Performance

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